

Lazy Closed Walks in Growing Dimension

All fixed orders, parity conditioned Poisson laws,
and occupied coordinate fluctuations for OEIS A328716

Part II: uniform lattice-bridge asymptotics, a Gaussian bound disproved,
a sharp dimension threshold, and sparse Poisson laws

Part I: Research report 179

Part II: research manuscript prepared for Vladimir Reshetnikov

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Abstract

Let a_n count the n -step walks from the origin to itself in $\mathbb{Z}^n$, with steps $0, \pm e_1, \dots, \pm e_n$. This is OEIS A328716. We prove its known parity-dependent leading equivalent, give exact Bessel definitions of its constants, and construct every fixed inverse-power correction, including three explicit terms. The marked expansion has an absolute complex-uniform remainder, so it remains valid when its leading amplitude vanishes.

For a uniformly chosen bridge, the zero-step count has an all-orders exponentially weighted ℓ^1 expansion about a parity-conditioned Poisson law. The number of occupied coordinate axes has Gaussian fluctuations, asymptotically independent of that residual Poisson variable, and an explicit parity-dependent constant mean correction. Separate inverse carriers retain both parity constants and give an eventual global integer bracket of width at most one. A compact proportional length-to-dimension extension retains the moving Poisson parameter.

The exact generating function and numerical leading equivalent were already recorded in the OEIS. Bessel walk enumeration and full large-powers expansions are classical. This is a family-specific derivation and refinement, with bounded source comparison, not a claim of an open-problem resolution or historical priority. Finite exact verification and optional high-precision diagnostics accompany the proofs; no effective inverse error constant or certified numerical starting threshold is claimed.

[write] **Status and trust boundary, added 7 October 2026, batch 109.** This is bundle Report 179 of an external AI-assisted research session, filed in ProveIt as the single-source research report **a328716-lazy-closed-walks** (provenance, the sources as the write read them, what was checked, relation to the repository, the collected non-claims and the reading conventions are in Section 1.1). It is unrefereed and not formalized. Its author line and PDF author field read “Research report 179”: it names no person or tool. *What rests on what.* Proposition 2.1, Lemma 3.1 and Theorems 4.1, 6.1, 7.1, 8.1 and 9.1 have conventional proofs in the text: the Bessel-product count, a two-arc saddle analysis with an absolute integrated remainder, Cauchy estimates on a larger disk, an analytic moving saddle with Lévy’s continuity theorem, and monotone parity brackets. The explicit $c_{\epsilon,2}$, $c_{\epsilon,3}$ come from the finite algorithm of Section 4, checked by the shipped programs; every decimal is a diagnostic. The exact formula and Kotěšovec’s numerical leading equivalent are in the OEIS, the Bessel walk generating function is Gessel–Weinstein–Wilf’s, and the large-powers method is classical (Flajolet–Sedgewick). The write adds Remarks 1.1, 8.2 and 9.2 (the OEIS entries quoted and the b-file compared; the inverses against the transseries volume; a proof of the row conjecture of A328718, a fixed-dimension statement outside the source’s regime) and dated notes. No statement, proof or number of the source was changed.

[write] **Part II, added 7 October 2026 (second write).** Since its second write the report has two Parts. Part I (Sections 1–12) is Report 179 as written and independently checked; the abstract and the status note above describe it alone, and nothing of it changed except the \part line, this note and dated notes at the ends of Sections 9 and 12. Part II (Sections 13–21, labels 1cw:ulb:) is the manuscript *Uniform Lattice-Bridge Asymptotics* (7 October 2026, “Prepared for Vladimir Reshetnikov”), which names Part I as its immediate predecessor and answers its Question 2 for the counts at both ends of the ratio N/D : a scalar saddle expansion uniform in the dimension and in all nonnegative activities, to every fixed order, with leading relative error at most $24/m$ (Theorem 16.2, Corollary 16.3); the dense Gaussian threshold $D^3/N^2 \rightarrow 0$ with critical factor $\exp(1/(48s^2))$ (Theorem 19.1); and, for $m^2/D = O(1)$, a joint Poisson law of repeated axes and parity-conditioned idle steps (Theorems 18.1 and 18.2). It also gives an exact counterexample, at $D = 16$ and $N = 26$, to the universal Gaussian return bound stated by Ball and Sterbenz as a conjecture of Lyons (Theorem 15.1), which the write confirmed by its own exact count. Section 13 records its provenance, what it answers here, the checks, its non-claims and its letters: they are not Part I’s (in Part II $\epsilon \in \{0, 1\}$ is the parity remainder of $N = 2m + \epsilon$; in Part I $\epsilon = (-1)^n$ is a sign).

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Part I

Lazy closed walks in growing dimension: all fixed orders, parity conditioned Poisson laws, and occupied coordinate fluctuations for OEIS A328716

1 The model and the scope of the results

A walk is an ordered sequence of steps from $\{0, +e_1, -e_1, \dots, +e_D, -e_D\}$, where the axes are labeled and the zero step is a single available step. A bridge has total displacement zero. Let

$$A_{N,D}(u) = \sum_{N\text{-step bridges in } \mathbb{Z}^D} u^{\text{number of zero steps}}, \quad A_n(u) = A_{n,n}(u), \quad a_n = A_n(1).$$

The empty bridge is counted once, including in dimension zero. Random bridges below are uniform over this finite set; they are equivalently uniform-step random walks conditioned to return to the origin. We write J_n for their zero-step count and K_n for the number of axes used at least once. An axis is counted once regardless of how many steps use it.

OEIS A328716 [1] records the exact diagonal formula, attributed to Ilya Gutkovskiy on 26 October 2019, and a leading equivalent with numerical parity constants, attributed to Václav Kotěšovec on 27 October 2019. The general array A328718 [2] identifies A328716 as its diagonal; its entry is due to Seiichi Manyama, and its walk interpretation is attributed to Alois P. Heinz. The classical Bessel-product walk generating function is given by Gessel, Weinstein, and Wilf [3], equation (2). Our lazy step appends an elementary exponential factor.

The large-powers method, including full descending-power expansions and compact-ratio uniformity, is classical; see Flajolet and Sedgewick [4], Theorem VIII.8 and its proof. Their subsequent remarks also discuss vanishing multipliers. We give complete arguments specialized to this period-two family, rather than interpreting the existence of a full expansion as a new analytic theorem. The explicit coefficients, marked probability refinements, occupation calculation, and inverse brackets are the objects of this report. No exhaustive historical novelty claim is made.

Every all-orders statement means a Poincaré expansion: the order M is fixed before the size parameter tends to infinity. Constants and onsets may depend on M and any specified compact sets. We do not assert convergence of the infinite series, allow M to grow, or supply computable remainder constants.

Remark 1.1 ([write] The OEIS entries, quoted, 7 October 2026). The entries were read on 7 October 2026 in the OEIS internal format; the quotations are verbatim.

- (a) A328716 (revision #38, 27 October 2019), offset 0, by Seiichi Manyama (26 October 2019): “Constant term in the expansion of $(1 + x_1 + x_2 + \dots + x_n + 1/x_1 + 1/x_2 + \dots + 1/x_n)^n$.”; a comment by Alois P. Heinz (26 October 2019) giving the walk interpretation; the formulas “ $a(n) = n! \cdot [x^n] \exp(x) \cdot \text{BesselI}(0, 2 \cdot x)^n$. - _Ilya Gutkovskiy_, Oct 26 2019” ((2) with $N = D = n, u = 1$) and “ $a(n) \sim c \cdot d^n \cdot n^n$, where $d = 0.8047104059195202206625458331930618795\dots$ and $c = 2.12946224998808159475495497\dots$ if n is even and $c = 1.4189559976544232606562785\dots$ if n is odd. - _Vaclav Kotesovec_, Oct 27 2019”; a b-file “Table of $n, a(n)$ for $n = 0..398$ ” (Manyama; terms 0–199 from Heinz).

- (b) A328718 (revision #35, 30 October 2019), the array $T(n, k)$ with n the dimension and k the number of steps, by Manyama, with Heinz’s walk comment and the comment “**Conjecture:** Row r is asymptotic to $(2*r+1)^{(n + r/2)} / (2^r * (Pi*n)^{(r/2)})$. - `_Vaclav Kotesovec_, Oct 27 2019`” (proved in Remark 9.2), and column polynomials $T(n, 2), \dots, T(n, 7)$ (Kotěšovec, 30 October 2019).

Kotěšovec’s d and both values of c agree to every printed digit with d , C_+ and C_- of (7), computed by the write at 60 digits (his decimals are truncations); Theorem 4.1 at $M = 0$ proves his equivalent, for which the entry gives no proof. The write compared all 399 b-file terms with the exact integer convolution of Section 10, and 44 of them ($n \leq 39$ and $n = 100, 199, 250, 398$) with a direct rational power of F , a different route: all agree, and a_n is strictly increasing from $n = 1$ with $a_0 = a_1$. The source’s twenty displayed terms agree. Nothing was submitted to the OEIS.

1.1 Provenance, sources and reading conventions

[write] **Added 7 October 2026, batch 109.** *Provenance.* This report is Report 179 of the session bundle that ProveIt’s docs/incoming intake processed as batch 109: the archive `Lazy_Closed_Walks_Asymptotics_and_Inverses_Source.zip` (617,713 bytes, SHA-256 `e433f40b68ba...e5473b44f613`, 26 files, no wrapper directory), arrival commit `60f54ea06`, placed in `f7e9e5c2f`. The text is the delivered `Report179.tex` (701 lines, dated 3 October 2026), shipped as `article.tex`. The core programs are shipped under `code/` (the root `build.py`, `guard_tests.py`, `test_build.py` and `verify_manifest.py` too), the optional ones as `code/optional-*.py`, the data and generated/ records under `data/`, `optional/README.md` as `optional-README.md`, and `README_CODE.md` and `SOURCE_AUDIT.md` at the root (the report `README` maps every name). Not shipped, and retrievable from the arrival commit: the delivered 16-page `Report179.pdf`, the checksum manifest `SHA256SUMS.json` (25 entries, verified at placement), and the delivered `README`, replaced by the report `README`. The package pins the ProveIt commit `83bef707` (3 October 2026), which exists; it carries no “prepared for private review” line and no personal data. It is a single-source report, so no merge choice was made. The write prefixed the 72 delivered labels with `lcw:` and updated the 65 references to them; it added this subsection, the notes marked [write] and Remarks 1.1, 8.2 and 9.2, each the last statement of its section, and inserted nothing before a delivered display, table or statement, so every theorem, section and equation of the source keeps its delivered number. The verified-prefix markers of Section 2, which `code/verify.py` reads, are unchanged.

The sources, as the write read them (7 October 2026).

- OEIS A328716 and A328718 in the internal format, and the A328716 b-file (Remark 1.1); the source read the official GitHub record of A328716 because the webpage did not load for it.
- The transseries volume `Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/transseries_and_inversion.tex`: `p0:def:three-inverses`, `p0:thm:staircase` (with `p0:rem:parity-instances`), `p0:prop:factorial-core`, `p0:thm:core-reversion`, `plt:def:lw-monomial-log-datum` and `plt:thm:lw-template`, read for Remark 8.2.
- Not read by the write: Gessel–Weinstein–Wilf [3] and Flajolet–Sedgewick [4]; the source’s account of them (equation (2); Theorem VIII.8 and Note VIII.37) is reported, not confirmed.

What was checked at intake. The batch-109 dossier read the manuscript and found no error; with its own exact polynomial power at $n = 120, 121$ it found r , d , $C_{\pm}$ as printed and the scaled residuals consistent with $c_{\epsilon,1}$, $c_{\epsilon,2}$; it confirmed the pin and the audit’s statements about the repository; and it reran the programs on copies: `code/verify.py` passing (its summary equal to `generated/verification.json` as JSON), `regenerate.py--compare` and both optional `--compare` checks exiting 0, `guard_tests.py` failing on Windows (text regenerated with CRLF). *What the write checked.* Every proof line by line, with no mathematical error found: the counting (2)–(3) and the injection; the Riccati equation (6) and $b = 4r^2 - 1$; $\kappa_3, \dots, \kappa_6$ from (8); the two-arc reduction and the parity of the odd coefficients; $\mathcal{T}_1$ and $c_{\epsilon,1}$ (21); the quotient and Cauchy steps of Theorem 6.1 and $Q_{\epsilon,1}$; Ψ ’s derivatives, τ^2 and the mean offset (33); the parity brackets and the width argument of Theorem 8.1; the closed form (42) and the displacement; and the compactness

argument of Theorem 9.1. With its own code (Python 3.14.4, mpmath 1.3.0): the constants of Sections 3 and 7 at 60 digits (note at the end of Section 3); $c_{\epsilon,1}, c_{\epsilon,2}, c_{\epsilon,3}$ from (21)–(23) against the table of Section 5; the diagnostics table of Section 10 to every printed digit; $\mathbb{E}K_n - np$ from exact marked counts; the b-file (Remark 1.1); and the row conjecture numerically (Remark 9.2). It reran the programs from the shipped files (README, Route B): `code/verify.py` passing with its summary equal to the shipped record as JSON, `regenerate.py--compare` and both optional checks exiting 0, `guard_tests.py` failing on Windows as at intake. Finite and floating computations are observations, not certificates.

Relation to the repository. Before batch 109 no file of the repository named A328716 or A328718. The source’s audit, pinned at 83befe707, read the A039831 report (now `a039831-two-fourier-peaks`, a moving maximum coefficient) and set aside A047909 and A108242 as covered (now `a047909-beta-renewal-subsequences` and `a108242-regular-cyclic-word-covers`) and Report 169’s packed matrices (now Part IV of `a261781-matrix-compositions`), all in `oeis-sequence-asymptotics/`; none shares a result with this report, so no reciprocal note is proposed. No statement of this report is formalized in Lean or Rocq, and its place in the collection confers no formal status. The inverses of Section 8 are compared with the transseries volume in Remark 8.2.

[write] **What is not claimed, collected from the source.** The exact formula and the numerical leading equivalent are already in the OEIS; the Bessel walk generating function and the large-powers method, including full expansions, compact-ratio uniformity and vanishing multipliers, are classical; no open-problem resolution, no historical priority (“bounded evidence, not an absence certificate”), and full priority for the individual probability refinements “remains unestablished”. Every expansion is a Poincaré expansion at fixed order: no convergence, no growing order, no computable remainder constants or onsets, no certified finite-input inverse calculator (“The existential width-one theorem is not by itself a certified finite-input inverse calculator”). The weighted expansion is a signed approximation; the occupation limits are weak limits, not total-variation convergence to a Gaussian. Nothing treats $\rho \downarrow 0$, $\rho \rightarrow \infty$ or fixed D , and A328718’s row statement is a separate regime. The diagnostics are not interval-certified; no OEIS record or external repository was edited. The write adds: its own checks are floating or finite; Remark 9.2 proves a fixed-dimension statement by the classical local limit argument, with no novelty claimed; Remark 8.2 claims no novelty for any inversion.

Reading conventions (letters the source reuses, with the tempting false readings; no symbol of the source was renamed; symbols of the transseries volume carry the subscript “vol” in Remark 8.2).

Symbol	Meaning here (where)	Not to be read as
$H(v, t), H(s),$ $H_\ell(j)$	the phase exponent (18); $h(e^s)$ (Section 7); the monomial polynomials (28)	each other
$h, h(w)$	a cumulant order or summation index; the moving log partition function (34)	each other; the volume’s h_{vol}
$R, R', R_2(n),$ $R_{k,m}$	disk radii (Theorems 4.1, 6.1); the scaled residual (Section 10); occupied-axis counts	each other
$q, Q_{D,k}, Q_{\epsilon,\ell}$	$1/f$ (Section 7), also the even index in (10); integer convolutions (45); probability polynomials	each other
K_n, K	occupied axes; a compact ratio interval (Section 9)	each other
$D, \mathcal{D}$	the dimension; $z d/dz$ (1)	each other
$\rho, r, r_\rho, r(w)$	N/D ; the saddle; the ratio saddle; the moving saddle	the volume’s ρ_{vol} (a residue class or a datum)
$\mathcal{T}_\ell, T(n, k)$	saddle functionals (11); the A328718 array (n the dimension)	each other
d, d_{TV}	$F(r)/(er)$; total variation	each other; the volume’s d_{vol}
$E_\epsilon, E_{\epsilon,u}, \mathbb{E}$	$e^t + \epsilon e^{-t}$; the marked amplitude; expectation	each other
v_ϵ, v	$rE'_\epsilon(r)/E_\epsilon(r)$ (also v in (21)–(23)); $n^{-1/2}$ in (18)	each other
s_j, s	Stirling coefficients (12); a real variable (Section 7)	each other
$\ell, \ell_{\epsilon,j}$	an expansion order; logarithmic coefficients (37)	each other
$A, A_n(u), A_M$	an amplitude; the counts; the inverse constant (39)	the volume’s a_{vol}
$F, \mathcal{F}_{\epsilon,M}$	$I_0(2z)$; the inverse carriers (38)	each other
$X, X_r, x, x(s)$	the inverse argument; the tilted variable (4); z^2 ; $\log r(e^s)$	each other; the volume’s X_{vol}
I, I_0, I_1	the indicator $\mathbf{1}_{\{X>0\}}$; Bessel functions	each other
κ_h, Λ, L_M	cumulants; $1 + \log(dt)$; the lower bracket	the volume’s $\kappa_{\text{vol}}, \Lambda_{\text{vol}}, L_{\text{vol}}$

2 Exact enumeration and tilted coordinates

Put

$$F(z) = I_0(2z) = \sum_{k \geq 0} \frac{z^{2k}}{(k!)^2}, \quad \mathcal{D} = z \frac{d}{dz}. \quad (1)$$

The defining series is sufficient for all arguments; no Bessel asymptotic formula is required.

Proposition 2.1. *For nonnegative integers N, D ,*

$$A_{N,D}(u) = N! [z^N] e^{uz} F(z)^D, \quad (2)$$

$$A_n(u, w) := \sum_{\text{bridges}} u^{J_n} w^{K_n} = n! [z^n] e^{uz} \{1 + w(F(z) - 1)\}^n. \quad (3)$$

In particular $J_n \equiv n \pmod{2}$, $0 \leq K_n \leq \lfloor (n - J_n)/2 \rfloor$, and $a_{n+1} > a_n$ for every $n \geq 1$.

Proof. If there are j zero steps and k_i positive and k_i negative steps in axis i , then $j + 2 \sum_i k_i = N$ and the number of orderings is

$$\frac{N!}{j! \prod_i (k_i!)^2}.$$

Summing proves (2); marking each nonzero k_i proves (3). The parity and support assertions follow from the same constraint. To prove strict monotonicity, embed a bridge in $\mathbb{Z}^{n+1}$ and append a zero

step. This is injective. A bridge using $+e_{n+1}$ once, $-e_{n+1}$ once, and $n - 1$ zero steps lies outside the image. The only initial equality is $a_0 = a_1 = 1$. $\square$

The first twenty values, beginning at $n = 0$, agree with the inspected OEIS record:

$$\begin{aligned} &1, 1, 5, 19, 217, 1451, 26041, 249705, 6116209, 76432627, \\ &2373097921, 36562658573, 1374991573825, 25188442156333, \\ &1112491608614933, 23620069750701091, 1198207214200181217, \\ &28930659427538020915, 1657461085278025906081, \\ &44848606508761385855085. \end{aligned}$$

For any $r > 0$ define an even-valued variable X_r by

$$\mathbb{P}(X_r = 2k) = \frac{r^{2k}}{(k!)^2 F(r)} \quad (k \geq 0). \quad (4)$$

Its cumulant generating function is $\log F(re^s) - \log F(r)$, so

$$\mathbb{E}X_r = \mathcal{D} \log F(r), \quad \text{Var } X_r = \mathcal{D}^2 \log F(r) > 0.$$

If $X_1, \dots, X_D$ have this law independently and $Z \sim \text{Poisson}(r)$ is independent, then, conditional on $Z + \sum_i X_i = N$, the pair $(Z, \sum_i \mathbf{1}_{\{X_i > 0\}})$ has exactly the zero-step and occupied-axis law of the uniform bridge. Indeed every admissible multiplicity vector has conditional mass proportional to $1/(j! \prod_i (k_i!)^2)$; its factor r^N cancels. This interpretation will explain the occupation variance.

3 The saddle and exact constants

Lemma 3.1. *There is a unique $r > 0$ such that*

$$\mathcal{D} \log F(r) = \frac{2r I_1(2r)}{I_0(2r)} = 1. \quad (5)$$

Writing $b = \mathcal{D}^2 \log F(r)$, one has $b = 4r^2 - 1 > 0$.

Proof. The mean in (4) is strictly increasing in $\log r$, since its derivative is the positive variance. It tends to zero as $r \downarrow 0$. It tends to infinity as $r \rightarrow \infty$: for any finite support cutoff, comparison with one higher positive term of F shows that the probability below the cutoff tends to zero. Thus the intermediate value theorem gives the unique solution.

Termwise differentiation of (1) gives $z^2 F'' + z F' - 4z^2 F = 0$. With $a(z) = \mathcal{D} \log F(z)$ this becomes

$$\mathcal{D}a = 4z^2 - a^2. \quad (6)$$

At $a(r) = 1$ it gives the asserted expression for b . $\square$

For $\epsilon \in \{+1, -1\}$ define

$$E_\epsilon(t) = e^t + \epsilon e^{-t}, \quad d = \frac{F(r)}{er}, \quad C_\epsilon = \frac{E_\epsilon(r)}{\sqrt{b}}, \quad v_\epsilon = \frac{r E'_\epsilon(r)}{E_\epsilon(r)}. \quad (7)$$

Both C_ϵ are positive. The signs $+1, -1$ always denote even and odd n , respectively. Thus $v_+ = r \tanh r$ and $v_- = r \coth r$. Useful numerical values are

$$\begin{aligned} r &= 0.804139735863439633472418655\dots, \\ b &= 1.586562859178089847374174939\dots, \\ d &= 0.804710405919520220662545833\dots, \\ C_+ &= 2.129462249988081594754954973\dots, \\ C_- &= 1.418955997654423260656278534\dots \end{aligned}$$

These reproduce the numerical leading constants already recorded in A328716. The decimal values are ordinary high-precision evaluations, not interval certificates.

All higher saddle cumulants have finite algebraic definitions. Set $x = z^2$, $a = \mathcal{D} \log F(z)$, and

$$\mathcal{L} = 2x\partial_x + (4x - a^2)\partial_a, \quad \kappa_h = \mathcal{L}^{h-1}a \Big|_{a=1, x=(b+1)/4}. \quad (8)$$

Then $\kappa_h = \mathcal{D}^h \log F(r)$ and, explicitly,

$$\begin{aligned} \kappa_1 &= 1, & \kappa_2 &= b, & \kappa_3 &= 2, \\ \kappa_4 &= -2b(b-2), & \kappa_5 &= 4(b-2)(b-1), \\ \kappa_6 &= 8(2b-3)(b^2-b+1). \end{aligned} \quad (9)$$

Equation (6) proves (8) inductively. In particular all cumulants are polynomials in b .

[write] **Rechecked, 7 October 2026.** Recomputed at 60 digits from (5): d , C_+ , C_- above and τ^2 in Section 7 are correct truncations, but the last printed digits of r , b and p are rounded up: the truncations are $r = 0.804139735863439633472418654\dots$ (next digits 99884), $b = 1.586562859178089847374174938\dots$ (92253) and, in Section 7, $p = 0.431494883273705551403382959\dots$ (59299). The delivered digits are kept; every printed digit except these three is a truncation. (Scope clarified after the independent check of 7 October 2026: this concerns the seven values printed with “ $\dots$ ”. The values printed without it, the six entries of the table of Section 5 and the two mean offsets of Theorem 7.1, are correctly rounded to twelve decimals, and five of them are not truncations.)

4 An absolute expansion to every fixed order

For an amplitude A analytic near r , define a linear functional $\mathcal{T}_\ell A$ as follows. Sum over $j \geq 0$ and finitely supported nonnegative integers $(m_h)_{h \geq 3}$ satisfying

$$j + \sum_{h \geq 3} (h-2)m_h = 2\ell, \quad q = j + \sum_{h \geq 3} hm_h. \quad (10)$$

The integer $q = 2\ell + 2 \sum_h m_h$ is even. Set

$$\mathcal{T}_\ell A = \sum_{(10)} \frac{(\mathcal{D}^j A)(r)}{j!} \frac{(-1)^{q/2} (q-1)!!}{b^{q/2}} \prod_{h \geq 3} \frac{\kappa_h^{m_h}}{m_h! (h!)^{m_h}}. \quad (11)$$

Here $(-1)!! = 1$. The sum is finite, uses amplitude derivatives only through 2ℓ , and cumulants only through $2\ell + 2$. In particular $\mathcal{T}_0 A = A(r)$.

Let s_j be the classical Stirling coefficients, specified to all fixed orders by the formal identity

$$\sum_{j \geq 0} s_j t^j = \exp \left(\sum_{k \geq 1} \frac{B_{2k}}{2k(2k-1)} t^{2k-1} \right), \quad \frac{x}{e^x - 1} = \sum_{j \geq 0} B_j \frac{x^j}{j!}. \quad (12)$$

Thus $s_0 = 1$, $s_1 = 1/12$, $s_2 = 1/288$, $s_3 = -139/51840$. Define

$$\mathcal{P}_\ell = \sum_{h=0}^{\ell} s_h \mathcal{T}_{\ell-h}, \quad E_{\epsilon,u}(z) = e^{uz} + \epsilon e^{-uz}. \quad (13)$$

All series used for coefficient extraction in this report are formal when an infinite expansion is displayed.

Theorem 4.1. *For each fixed $M \geq 0$ and $R < \infty$, uniformly for complex $|u| \leq R$ as $n \rightarrow \infty$,*

$$A_n(u) = \frac{(dn)^n}{\sqrt{b}} \left\{ \sum_{\ell=0}^M \frac{\mathcal{P}_\ell E_{\epsilon,u}}{n^\ell} + O_{M,R}(n^{-M-1}) \right\}, \quad \epsilon = (-1)^n. \quad (14)$$

The error inside braces is absolute, including at zeros of $E_\epsilon(ru)$. In particular, with $c_{\epsilon,\ell} = \mathcal{P}_\ell E_{\epsilon,1}/E_\epsilon(r)$,

$$a_n = C_\epsilon (dn)^n \left\{ \sum_{\ell=0}^M \frac{c_{\epsilon,\ell}}{n^\ell} + O_M(n^{-M-1}) \right\}, \quad c_{\epsilon,0} = 1. \quad (15)$$

4.1 The two dominant arcs

Cauchy's formula on $|z| = r$ gives

$$[z^n] e^{uz} F(z)^n = \frac{F(r)^n r^{-n}}{2\pi} \int_{-\pi}^{\pi} e^{ure^{i\theta}} \left(\frac{F(re^{i\theta})}{F(r)} \right)^n e^{-in\theta} d\theta. \quad (16)$$

Equality in $|F(re^{i\theta})| \leq F(r)$ requires $e^{2ik\theta} = 1$ for every $k \geq 0$, since every even coefficient is strictly positive and the constant coefficient is nonzero. Hence it occurs only at $\theta = 0, \pi$ modulo 2π . Outside fixed small neighborhoods of these points the ratio is bounded by a constant $\gamma < 1$. The amplitude is uniformly bounded for $|u| \leq R$, so those arcs are exponentially small.

Near zero an analytic logarithm exists and

$$\phi(\theta) := \log \frac{F(re^{i\theta})}{F(r)} - i\theta = -\frac{b\theta^2}{2} + \sum_{h \geq 3} \frac{\kappa_h (i\theta)^h}{h!}. \quad (17)$$

By shrinking the neighborhood, $\Re \phi(\theta) \leq -b\theta^2/4$. Near π , translating the angle by π gives the identical phase and amplitude $\epsilon e^{-ure^{i\theta}}$. Thus the two local integrals combine into a single symmetric integral with amplitude $E_{\epsilon,u}(re^{i\theta})$.

4.2 A controlled integrated remainder

Fix any $0 < \delta < 1/2$ and restrict further to $|\theta| \leq n^{-1/2+\delta}$. The removed intermediate arcs contribute $O(\exp(-cn^{2\delta}))$ relative to the saddle scale, even after multiplication by any fixed power of n . Put $t = \sqrt{n}\theta$ and $v = n^{-1/2}$. After extracting $e^{-bt^2/2}$, the remaining phase exponent is

$$H(v, t) = \sum_{h \geq 3} \frac{\kappa_h (it)^h v^{h-2}}{h!}, \quad (18)$$

an analytic function of v at zero. On $|t| \leq n^\delta$, uniformly for $0 \leq v' \leq v$, convergence of the phase series yields

$$|H(v', t)| \leq Cv|t|^3 \leq Cn^{-1/2+\delta}t^2.$$

For all sufficiently large n the last constant is smaller than, for example, $b/8$. Each fixed derivative of H in v' is bounded by a fixed polynomial in $|t|$, uniformly on this set. This follows by differentiating its convergent series with $|v't|$ in a fixed sufficiently small disk. Likewise the k th v' derivative of $A(re^{iv't})$ is bounded by $C_k|t|^k$ for amplitudes in any compact analytic family.

The product $A(re^{iv't})e^{H(v',t)}$ therefore has every fixed v' derivative bounded by a polynomial in $|t|$ times $e^{bt^2/8}$. Taylor's formula through degree $2M+1$ in v has, after multiplication by $e^{-bt^2/2}$ and integration, a remainder

$$O(v^{2M+2}) \int_{\mathbb{R}} (1+|t|)^L e^{-3bt^2/8} dt = O(n^{-M-1}).$$

This bound is uniform for $A = E_{\epsilon,u}$, $|u| \leq R$. A coefficient of odd degree in v is an odd polynomial in t : in the notation (10), its total t degree differs from its v degree by an even integer. Such coefficients integrate to zero on the symmetric interval. Replacing each remaining finite-range Gaussian moment by the full moment has an exponentially small error.

The normalized Gaussian moments are

$$\frac{\int_{\mathbb{R}} (it)^q e^{-bt^2/2} dt}{\int_{\mathbb{R}} e^{-bt^2/2} dt} = (-1)^{q/2} (q-1)!! b^{-q/2} \quad (q \text{ even}).$$

Expanding the amplitude and the exponential in (18) now gives precisely (11), with coefficient constraint (10). We have proved

$$[z^n] e^{uz} F(z)^n = \frac{F(r)^n r^{-n}}{\sqrt{2\pi n b}} \left\{ \sum_{\ell=0}^M \frac{\mathcal{T}_{\ell} E_{\epsilon,u}}{n^{\ell}} + O_{M,R}(n^{-M-1}) \right\}. \quad (19)$$

Finally use the fixed-order Stirling expansion $n! = \sqrt{2\pi n} (n/e)^n (\sum_{j=0}^M s_j n^{-j} + O_M(n^{-M-1}))$. Multiplying gives (14). Since $E_{\epsilon}(r) > 0$, dividing only at $u = 1$ proves (15). This completes the proof of Theorem 4.1.

Remark 4.2. For even n , $u = i\pi/(2r)$ makes $E_+(ru) = 0$; for odd n , $u = i\pi/r$ makes $E_-(ru) = 0$. Formula (14) still controls the first nonzero correction there. A relative error of the form $E_{\epsilon}(ru)(1 + O(n^{-1}))$ would not give this statement and generally would be false at these points. No division by a marked leading amplitude occurs in the proof.

5 The first three explicit corrections

The normalized amplitude derivatives are the moments of the limiting parity law that will appear in Section 6. Direct differentiation gives

$$\mu_j := \frac{(\mathcal{D}^j E_{\epsilon,1})(r)}{E_{\epsilon}(r)} = \sum_{h=0}^j \left\{ \begin{matrix} j \\ h \end{matrix} \right\} \left(\frac{b+1}{4} \right)^{[h/2]} \begin{cases} 1, & h \text{ even}, \\ v_{\epsilon}, & h \text{ odd}, \end{cases} \quad (20)$$

with $\mu_0 = 1$ and Stirling numbers of the second kind in braces. This follows from the Touchard identity $(z\partial_z)^j e^z = e^z \sum_h \left\{ \begin{matrix} j \\ h \end{matrix} \right\} z^h$ and $r^2 = (b+1)/4$.

Equations (8), (11), (12), and (20) constitute a finite exact algorithm at arbitrary fixed order. In the following formulas put $v = v_{\epsilon}$:

$$c_{\epsilon,1} = \frac{9-7b}{24b} - \frac{5}{6b^3} + \frac{(2-b)v}{2b^2}. \quad (21)$$

For readability write the next two as polynomials divided by their common denominator:

$$\begin{aligned}
 1152b^6 c_{\epsilon,2} &= 193b^6 + 312b^5v - 594b^5 - 2856b^4v \\
 &\quad + 1245b^4 + 4368b^3v + 1624b^3 + 3360b^2v \\
 &\quad - 6552b^2 - 6720bv + 6160,
 \end{aligned} \tag{22}$$

$$\begin{aligned}
 414720b^9 c_{\epsilon,3} &= -37427b^9 + 105660b^8v + 222507b^8 \\
 &\quad + 934560b^7v - 1371537b^7 - 8147700b^6v \\
 &\quad + 2306661b^6 + 8541000b^5v + 8767656b^5 \\
 &\quad + 23476320b^4v - 22863852b^4 - 36146880b^3v \\
 &\quad - 9517200b^3 - 14414400b^2v + 45405360b^2 \\
 &\quad + 28828800bv - 27227200.
 \end{aligned} \tag{23}$$

They evaluate as follows:

Parity	$c_{\epsilon,1}$	$c_{\epsilon,2}$	$c_{\epsilon,3}$
Even	-0.219966033027	0.070696999974	-0.011375589727
Odd	-0.164864931187	0.080945110173	-0.071114538915

For example, the first saddle functional is

$$\mathcal{T}_1 A = -\frac{\mathcal{D}^2 A(r)}{2b} + \frac{\kappa_3 \mathcal{D} A(r)}{2b^2} + \left(\frac{\kappa_4}{8b^2} - \frac{5\kappa_3^2}{24b^3} \right) A(r). \tag{24}$$

Adding the Stirling term $A(r)/12$, substituting (9), and using $\mu_1 = v$, $\mu_2 = r^2 + v$ yields (21). The finite algorithm gives (22)–(23); an independent formal-exponential recurrence provides an algebraic cross-check described in Section 10.

6 Zero steps and weighted probability expansions

Let J_ϵ be Poisson(r) conditioned on $(-1)^{J_\epsilon} = \epsilon$. Write its mass function and probability generating function as

$$\pi_\epsilon(j) = \begin{cases} \frac{2r^j}{j!E_\epsilon(r)}, & (-1)^j = \epsilon, \\ 0, & \text{otherwise,} \end{cases} \quad \mathbb{E}u^{J_\epsilon} = \frac{E_\epsilon(ru)}{E_\epsilon(r)}. \tag{25}$$

In particular $\mathbb{E}J_\epsilon = v_\epsilon$ and $\mathbb{E}J_\epsilon^2 = r^2 + v_\epsilon$.

Theorem 6.1. *For every fixed $M \geq 0$ and $R > 1$, there are polynomials $Q_{\epsilon,\ell}(j)$ of degree at most 2ℓ , with $Q_{\epsilon,0} = 1$, such that*

$$\sum_{j \geq 0} R^j \left| \mathbb{P}(J_n = j) - \pi_\epsilon(j) \sum_{\ell=0}^M \frac{Q_{\epsilon,\ell}(j)}{n^\ell} \right| = O_{M,R}(n^{-M-1}), \quad \epsilon = (-1)^n. \tag{26}$$

For $\ell \geq 1$, $\mathbb{E}Q_{\epsilon,\ell}(J_\epsilon) = 0$. The first polynomial is

$$Q_{\epsilon,1}(j) = -\frac{j^2 - r^2 - v_\epsilon}{2b} + \frac{j - v_\epsilon}{b^2}. \tag{27}$$

Consequently the total-variation distance from J_ϵ is $O(n^{-1})$, and every fixed polynomial moment has an expansion to all fixed orders.

Proof. Choose $R' > R$. Theorem 4.1 holds uniformly on $|u| \leq R'$. Dividing by the expansion at $u = 1$ is legitimate since its leading coefficient is the positive $E_\epsilon(r)$. Finite series division gives a probability-generating-function expansion with an absolute remainder uniformly $O(n^{-M-1})$ on that disk.

To identify its coefficients explicitly, define the monomial polynomials

$$H_\ell(j) = r^{-j} \mathcal{P}_\ell(z^j). \quad (28)$$

Since $\mathcal{D}^k z^j = j^k z^j$, these are polynomials in j of degree at most 2ℓ , and

$$c_{\epsilon,\ell} = \mathbb{E} H_\ell(J_\epsilon).$$

The probability polynomials are specified by the finite recurrence

$$Q_{\epsilon,0} = 1, \quad Q_{\epsilon,\ell}(j) = H_\ell(j) - \sum_{h=1}^{\ell} c_{\epsilon,h} Q_{\epsilon,\ell-h}(j). \quad (29)$$

Indeed the coefficient of u^j in $\mathcal{P}_\ell E_{\epsilon,u}$ is $(1 + \epsilon(-1)^j) r^j H_\ell(j)/j!$. Dividing by the unmarked denominator gives (29), including exact zeros at the wrong parity. Taking expectation in that recurrence proves centering inductively.

If $\mathcal{R}_n(u)$ denotes the analytic remainder after quotient expansion, Cauchy's inequality gives

$$|[u^j] \mathcal{R}_n(u)| \leq C n^{-M-1} (R')^{-j}.$$

Multiplication by R^j and summation of the geometric series proves (26). This step controls the infinite tail; pointwise coefficient convergence alone would not suffice.

Finally (24), $\kappa_3 = 2$, and the subtraction of the mean in (29) give (27). Total variation is one half the unweighted ℓ^1 norm. Any fixed polynomial is bounded in absolute value by a constant times R^j , giving the moment statement. $\square$

The finite correction in (26) is a signed approximation; it is not asserted to be a probability mass function term by term. Setting $M = 0$ shows $J_n \Rightarrow J_\epsilon$ along each parity subsequence. The whole sequence has no single zero-step weak limit, since even and odd support remain disjoint. In particular the zero steps stay of order one even though the walk length and dimension both grow.

7 Occupied axes and a Gaussian product limit

Put

$$f = F(r), \quad q = f^{-1}, \quad p = 1 - q, \quad \tau^2 = pq - \frac{q^2}{b}. \quad (30)$$

Numerically,

$$p = 0.431494883273705551403382960\dots, \quad \tau^2 = 0.041597460122151058557181923\dots$$

Theorem 7.1. *One has $\tau^2 > 0$ and*

$$\frac{K_n - np}{\sqrt{n}} \Rightarrow \mathcal{N}(0, \tau^2). \quad (31)$$

Along a fixed parity $\epsilon = (-1)^n$, jointly

$$\left(\frac{K_n - np}{\sqrt{n}}, J_n \right) \Rightarrow (G, J_\epsilon), \quad (32)$$

where $G \sim \mathcal{N}(0, \tau^2)$ is independent of J_ϵ . Moreover

$$\mathbb{E}K_n = np - \frac{q}{b} \left(v_\epsilon + \frac{b-1}{2} - \frac{1}{b} \right) + O(n^{-1}), \quad \text{Var } K_n = n\tau^2 + O(1). \quad (33)$$

The constant mean offsets are approximately -0.071243028912 for even n and -0.311664057977 for odd n .

7.1 An analytic moving saddle

Let

$$B_w(z) = 1 + w(F(z) - 1).$$

Near $w = 1$, the equation $\mathcal{D} \log B_w(r(w)) = 1$ has an analytic solution with $r(1) = r$. The implicit-function theorem applies in the variable $\log r$, because the derivative is $b > 0$. Shrink to a fixed complex neighborhood of $w = 1$ in which $r(w)$, $B_w(r(w))$, and

$$b(w) = \mathcal{D}^2 \log B_w(r(w))$$

are nonzero and $\Re b(w)$ is bounded below by a positive constant. Take logarithm and square-root branches from their real positive values at $w = 1$, and put

$$h(w) = \log B_w(r(w)) - \log r(w). \quad (34)$$

For completeness the moving contour is valid also for complex w . The parametrization $z = r(w)e^{i\theta}$ describes a positively oriented circle of radius $|r(w)|$, merely with a rotated starting point. The coefficient integrand is entire except for its pole at zero, so Cauchy's formula applies on this circle. At $\theta = 0$ the exact saddle equation cancels the linear term and gives quadratic term $-b(w)\theta^2/2$. Since B_w is even, the second saddle is exactly $-r(w)$ and its phase factor is still ϵ .

At $w = 1$ the compact complement of the two local arcs has a strict normalized modulus gap. Continuity preserves it uniformly for all complex w in a sufficiently small closed neighborhood of one. Local logarithms exist on the two arcs; a global logarithm is not needed. The integrated Taylor-remainder argument of Section 4.2 then holds with complex analytic cumulants, uniformly in w and bounded u . The Gaussian integral is $\sqrt{2\pi/b(w)}$ with the specified branch, by analytic continuation in the right half-plane or direct Gaussian integration. In particular,

$$A_n(u, w) = \left(\frac{n}{e} \right)^n \frac{e^{nh(w)}}{\sqrt{b(w)}} \left\{ E_\epsilon(ur(w)) + O(n^{-1}) \right\}. \quad (35)$$

The remainder inside braces is absolute and analytic in u, w . The full fixed-order version follows in the same way. Cauchy's derivative estimate on a smaller w disk allows its uniform remainders to be differentiated any fixed number of times.

7.2 The variance density and limiting independence

Write $H(s) = h(e^s)$ and $x(s) = \log r(e^s)$. At $s = 0$ consider the pair (X, I) , where $X = X_r$ and $I = \mathbf{1}_{\{X > 0\}}$. The log partition function is

$$\Psi(s, x) = \log(1 + e^s(F(e^x) - 1)).$$

The saddle constraint is $\Psi_x(s, x(s)) = 1$. At $(0, \log r)$ its derivatives are

$$\Psi_s = p, \quad \Psi_{ss} = pq, \quad \Psi_{xx} = b, \quad \Psi_{sx} = \text{Cov}(I, X) = q.$$

The covariance identity follows from $IX = X$ and $\mathbb{E}X = 1$. Differentiating the constraint and $H(s) = \Psi(s, x(s)) - x(s)$ gives

$$x'(0) = -\frac{q}{b}, \quad H'(0) = p, \quad H''(0) = pq - \frac{q^2}{b} = \tau^2. \quad (36)$$

Strict Cauchy–Schwarz makes $\tau^2 > 0$, since I is not an affine function of X on its positive-probability support $0, 2, 4, \dots$

For real t, y , put $w = e^{it/\sqrt{n}}$ and $u = e^{iy}$ in (35). Divide by $A_n(1, 1)$ and multiply by $e^{-itp\sqrt{n}}$. Taylor expansion gives

$$n\{H(it/\sqrt{n}) - H(0)\} - itp\sqrt{n} = -\frac{\tau^2 t^2}{2} + O(n^{-1/2}).$$

The remaining amplitude tends to $E_\epsilon(re^{iy})/E_\epsilon(r)$, including at its zeros because the remainder was absolute. The joint characteristic functions therefore tend to

$$e^{-\tau^2 t^2/2} \frac{E_\epsilon(re^{iy})}{E_\epsilon(r)}.$$

This is the characteristic function of the independent product in (32); Lévy’s continuity theorem applies. Setting $y = 0$ proves (31) without a parity restriction. These are weak limits, not total-variation convergence from a lattice variable to a continuous Gaussian.

7.3 The constant mean correction

At $u = 1$ the leading amplitude in (35) is nonzero in a small w neighborhood. Its analytic logarithm yields

$$\log \frac{A_n(1, e^s)}{A_n(1, 1)} = n\{H(s) - H(0)\} + \log \frac{E_\epsilon(r(e^s))}{E_\epsilon(r)} - \frac{1}{2} \log \frac{b(e^s)}{b} + O(n^{-1}).$$

The remainder may be differentiated, as established above. At fixed x , differentiating Ψ_{xx} with respect to s gives $\Psi_{xss} = q(b-1)$. One direct check is $\mathcal{D}p = q$ and $\mathcal{D}^2p = q(b-1)$ at the saddle. Allowing x to move then gives, using $\kappa_3 = 2$,

$$\left. \frac{d}{ds} b(e^s) \right|_{s=0} = q(b-1) + \kappa_3 x'(0) = q \left(b - 1 - \frac{2}{b} \right).$$

The derivative of $\log E_\epsilon(r(e^s))$ is $v_\epsilon x'(0)$. Together with (36) these prove the expectation formula in (33). Two derivatives prove the variance statement. The same analytic expansion supplies all fixed cumulant corrections, although only the first mean offset is needed here.

8 Parity sensitive inversion

Let

$$\nu(X) = \min\{n \geq 0 : a_n \geq X\}, \quad X \rightarrow \infty.$$

Strict monotonicity from Proposition 2.1 makes this a genuine threshold. For each parity let $\nu_\epsilon(X)$ be the first such integer of parity ϵ , so $\nu(X) = \min_\epsilon \nu_\epsilon(X)$.

Define the logarithmic coefficients formally by

$$\ell_{\epsilon,j} = [t^j] \log \left(\sum_{k \geq 0} c_{\epsilon,k} t^k \right). \quad (37)$$

For example $\ell_{\epsilon,1} = c_{\epsilon,1}$, $\ell_{\epsilon,2} = c_{\epsilon,2} - c_{\epsilon,1}^2/2$, and $\ell_{\epsilon,3} = c_{\epsilon,3} - c_{\epsilon,1}c_{\epsilon,2} + c_{\epsilon,1}^3/3$. For fixed $M \geq 0$ put

$$\mathcal{F}_{\epsilon,M}(t) = t \log(dt) + \log C_{\epsilon} + \sum_{j=1}^M \ell_{\epsilon,j} t^{-j}. \quad (38)$$

It is eventually increasing, with derivative $1 + \log(dt) + O_M(t^{-2})$, and tends to infinity. For large X let $t_{\epsilon,M}(X)$ be its unique sufficiently large solution of $\mathcal{F}_{\epsilon,M}(t) = \log X$. Write $\lceil t \rceil_{\epsilon}$ for the least integer at least t with $(-1)^n = \epsilon$.

Theorem 8.1. *For each fixed M there exists $A_M > 0$ such that, for all sufficiently large X , with*

$$\eta_{\epsilon} = \frac{A_M t_{\epsilon,M}^{-M-1}}{1 + \log(dt_{\epsilon,M})}, \quad (39)$$

one has

$$\lceil t_{\epsilon,M} - \eta_{\epsilon} \rceil_{\epsilon} \leq \nu_{\epsilon}(X) \leq \lceil t_{\epsilon,M} + \eta_{\epsilon} \rceil_{\epsilon}. \quad (40)$$

Taking the minimum of the two lower endpoints and of the two upper endpoints gives integers $L_M(X), U_M(X)$ satisfying

$$L_M(X) \leq \nu(X) \leq U_M(X), \quad U_M(X) - L_M(X) \in \{0, 1\} \quad (41)$$

eventually. No effective value of A_M or onset is asserted.

Proof. Taking logarithms in (15) gives, on each parity,

$$\log a_n = \mathcal{F}_{\epsilon,M}(n) + O_M(n^{-M-1}).$$

The leading relative factor is positive and tends to one, so the logarithm and its fixed-order remainder are valid. On a fixed-width neighborhood of $t = t_{\epsilon,M}$ the carrier derivative is comparable to $\Lambda = 1 + \log(dt)$, and n^{-M-1} is comparable to t^{-M-1} . Choose A_M strictly larger than the common logarithmic error bound after these comparisons.

The parity integer immediately preceding $\lceil t - \eta_{\epsilon} \rceil_{\epsilon}$ is below $t - \eta_{\epsilon}$ and within a fixed distance of t . By the mean value theorem its carrier is smaller than $\log X$ by more than the possible logarithmic error, so its actual count is below X . The integer $\lceil t + \eta_{\epsilon} \rceil_{\epsilon}$ has count at least X by the same argument with opposite sign. Monotonicity proves (40), including endpoint cases. Taking minima is valid because the global threshold is the minimum of the two parity thresholds.

It remains to prove the width assertion. At a common large argument the two carriers differ by $\log(C_+/C_-) + O_M(t^{-1})$. Their derivatives are of order $\log t$, so their roots differ by $O(1/\log t)$; their radii tend to zero. The convex hull of both real intervals $[t_{\epsilon,M} - \eta_{\epsilon}, t_{\epsilon,M} + \eta_{\epsilon}]$ therefore eventually has length less than one.

If $U_M > L_M$, choose a parity attaining the minimum lower ceiling L_M . Its upper ceiling must exceed L_M , so its real interval contains the integer L_M and extends above it. Every point of the other real interval is strictly below $L_M + 1$, since the common hull has length less than one. The opposite-parity upper ceiling is consequently at most $L_M + 1$. Thus $U_M \leq L_M + 1$. The alternative is $U_M = L_M$, completing (41). $\square$

For $M = 0$ the carriers have a closed form. With W_0 the positive real Lambert branch and $Y_{\epsilon} = \log(X/C_{\epsilon})$,

$$t_{\epsilon,0}(X) = \frac{Y_{\epsilon}}{W_0(dY_{\epsilon})}. \quad (42)$$

Substitution verifies the equation exactly. If instead one starts from $t_* = (\log X)/W_0(d \log X)$, the first displacement is

$$t_{\epsilon,0} - t_* = -\frac{\log C_\epsilon}{1 + \log(dt_*)} + o\left(\frac{1}{\log t_*}\right).$$

The two constants cannot be silently merged for sharp parity rounding. Newton iteration on (38) computes higher real carriers; interval rounding still requires (40). A ceiling is certified only when its two parity endpoints coincide, or the global endpoints coincide. The existential width-one theorem is not by itself a certified finite-input inverse calculator.

Remark 8.2 ([write] The inversions and the transseries volume, 7 October 2026). Compared, statement by statement, with the transseries volume (`Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/transseries_and_inversion.tex`); symbols of the volume carry the subscript “vol”.

- (a) *Instance.* The threshold $\nu(X)$ is the integer staircase N_* of `p0:def:three-inverses` with $A_n = a_n$ and $n_1 = 1$ (strictly increasing by Proposition 2.1), for $X > 1$, and the decomposition $\nu = \min_\epsilon \nu_\epsilon$ is `p0:thm:staircase(4)` with $r_{\text{vol}} = 2$: the parity ceiling $\lceil t \rceil_\epsilon$ is $\rho_{\text{vol}} + 2\lceil (t - \rho_{\text{vol}})/2 \rceil$ for the residue $\rho_{\text{vol}} \in \{0, 1\}$ with $(-1)^{\rho_{\text{vol}}} = \epsilon$, as in `p0:rem:parity-instances`, and each ν_ϵ is the parity ceiling of the inverse of any admissible interpolation of its class.
- (b) *Theorem 8.1: the parity brackets (40) are an analogue as stated and proved; also a consequence of `p0:thm:staircase(4)` once interpolations are supplied.* The source brackets each ν_ϵ directly about the root $t_{\epsilon,M}$ of the carrier, which is not the inverse of an interpolation of the counts; inside each class the conclusion has the form of `p0:thm:staircase(2)`. A second route (added by the write): with $r_n = \log a_n - \mathcal{F}_{\epsilon,M}(n) = O(n^{-M-1})$ on the class, let $\tilde{r}_\epsilon$ interpolate (r_n) linearly between consecutive class members and $\mathcal{A}_\epsilon = \mathcal{F}_{\epsilon,M} + \tilde{r}_\epsilon$. Since $\mathcal{F}'_{\epsilon,M} \asymp \log t$ and $|\tilde{r}'_\epsilon| \leq \max |r_n|$, $\mathcal{A}_\epsilon$ is eventually continuous and strictly increasing with $\mathcal{A}_\epsilon(n) = \log a_n$ on the class, so it is admissible there; for its inverse ν^ϵ at $\log X$, $|\nu^\epsilon - t_{\epsilon,M}| \leq |\tilde{r}_\epsilon(\nu^\epsilon)|/\min \mathcal{F}'_{\epsilon,M} = O(t^{-M-1}/\log t)$, at most η_ϵ for a suitable A_M , and the monotonicity of $\lceil \cdot \rceil_\epsilon$ gives (40). The width statement (41) is the source’s own; the volume asserts nothing of the kind.
- (c) *Instance.* The closed form (42) is `p0:prop:factorial-core` with $\kappa_{\text{vol}} = 1$, $d_{\text{vol}} = \log d$ and $L_{\text{vol}} = Y_\epsilon$: there $v = W_0(dY_\epsilon)$ and $X_{\text{vol}} = e^v/d = Y_\epsilon/v$, with the side condition $dt > 1$; the displacement $t_{\epsilon,0} - t_*$ is the first-order term given by the core’s slope $1/(\kappa_{\text{vol}}(v+1)) = 1/(1 + \log(dt))$. Its large- Y expansion is a *formal* instance of `plt:thm:lw-template` with $Z_{\text{vol}} = \log t$, $\xi_{\text{vol}} = \log Y_\epsilon$ and data $(\rho_{\text{vol}}, \mu_{\text{vol}}, \beta_{\text{vol}}, B_{\text{vol}}) = (1, 1, 0, \log(1 + t_{\text{vol}} \log d))$, since $\log(t \log(dt)) = Z_{\text{vol}} + \log Z_{\text{vol}} + \log(1 + \log d/Z_{\text{vol}})$.
- (d) *Formal instance after a change of variable.* For $M \geq 1$, with $t = t_{\epsilon,0}(1 + E)$, $t_{\text{vol}} = 1/t_{\epsilon,0}$ and $L = \log(dt_{\epsilon,0})$, the carrier equation $\mathcal{F}_{\epsilon,M}(t) = \log X$ divided by $t_{\epsilon,0}L$ reads $\Lambda_{\text{vol}}E + h_{\text{vol}}(E) + f_{\text{vol}}(t_{\text{vol}}, E) = 0$ with $\Lambda_{\text{vol}} = 1 + 1/L$, $h_{\text{vol}}(u) = \{(1+u)\log(1+u) - u\}/L$ and $f_{\text{vol}} = \sum_{j=1}^M \ell_{\epsilon,j} t_{\text{vol}}^{j+1} (1+u)^{-j}/L$, which is `p0:thm:core-reversion` over the ring generated over $\mathbb{Q}$ by the $\ell_{\epsilon,j}$, L^{-1} and $(1+L)^{-1}$. The source computes the carriers by Newton iteration instead and states no such series.
- (e) *Not shown to be an instance.* For $M \geq 1$ the full carrier equation is not shown to be an instance of `plt:thm:lw-template`: in the chart of (c) the terms $\ell_{\epsilon,j} t^{-j}$ are of relative size $e^{-(j+1)Z_{\text{vol}}}$, not a series in $t_{\text{vol}} = 1/Z_{\text{vol}}$ with coefficients polynomial in $\log Z_{\text{vol}}$, so (H3) fails there. The write does not claim that no other chart makes it an instance.

No novelty is claimed for any of these inversions.

9 Proportional length and dimension

Let $D \rightarrow \infty$ and N be an integer such that $\rho = N/D$ belongs to a fixed compact interval $K \subset (0, \infty)$. Define $r_\rho > 0$ by

$$\mathcal{D} \log F(r_\rho) = \rho, \quad b_\rho = \mathcal{D}^2 \log F(r_\rho) > 0, \quad \kappa_{\rho,h} = \mathcal{D}^h \log F(r_\rho).$$

The argument of Lemma 3.1 proves existence and uniqueness. Compactness of K puts the saddles in a compact positive interval and bounds b_ρ away from zero. Let $\mathcal{T}_{\rho,\ell}$ denote (11) with r, b, κ_h replaced by these quantities.

Theorem 9.1. *For fixed $M \geq 0$, $R < \infty$, uniformly for $\rho \in K$ and complex $|u| \leq R$,*

$$A_{N,D}(u) = \frac{N! F(r_\rho)^D r_\rho^{-N}}{\sqrt{2\pi D b_\rho}} \left\{ \sum_{\ell=0}^M \frac{\mathcal{T}_{\rho,\ell} E_{\epsilon,u}}{D^\ell} + O_{M,R,K}(D^{-M-1}) \right\}, \quad \epsilon = (-1)^N. \quad (43)$$

The remainder is absolute inside braces. Here $E_{\epsilon,u}(z) = e^{uz} + \epsilon e^{-uz}$ as before; $N!$ is retained exactly.

Let $J_{N,D}$ count zero steps. It has an all-fixed-orders weighted ℓ^1 expansion, uniform for $\rho \in K$, about $\text{Poisson}(r_\rho)$ conditioned to have parity N . In particular, for every fixed $R > 1$,

$$\sum_{j \geq 0} R^j |\mathbb{P}(J_{N,D} = j) - \pi_{\rho,\epsilon}(j)| = O_{R,K}(D^{-1}), \quad (44)$$

where $\pi_{\rho,\epsilon}$ is (25) with r replaced by r_ρ . If additionally $\rho \rightarrow \rho_0 \in (0, \infty)$ and N has fixed parity ϵ , then $J_{N,D}$ converges weakly to $\text{Poisson}(r_{\rho_0})$ conditioned on that parity.

Proof. Use the circle $|z| = r_\rho$ and centered phase

$$\log \frac{F(r_\rho e^{i\theta})}{F(r_\rho)} - i\rho\theta.$$

The quadratic coefficient is $-b_\rho/2$, uniformly negative. Positivity and the even support give exactly two maximum-modulus points for every ρ ; compactness supplies a uniform strict gap away from the two fixed neighborhoods. Local analytic derivatives and amplitude bounds are uniform on compact sets. The argument of Section 4.2, scaled by $\sqrt{D}$, therefore proves (43). Translation through π contributes $e^{-iN\pi} = (-1)^N$, independent of the parity of D .

The unmarked leading amplitudes $E_\epsilon(r_\rho)$ are bounded away from zero for both signs on K . Quotient expansion and the larger-disk Cauchy argument of Theorem 6.1 prove (44) and all its fixed-order refinements. More explicitly set

$$H_{\rho,\ell}(j) = r_\rho^{-j} \mathcal{T}_{\rho,\ell}(z^j), \quad c_{\rho,\epsilon,\ell} = \mathbb{E} H_{\rho,\ell}(J_{\rho,\epsilon}),$$

where $J_{\rho,\epsilon}$ has mass $\pi_{\rho,\epsilon}$, and apply the recurrence (29) with these quantities. The resulting polynomials of degree at most 2ℓ give an error $O_{M,R,K}(D^{-M-1})$. Finally continuity of r_ρ gives the asserted fixed-law weak limit when ρ converges and parity is fixed. $\square$

If desired, expanding $N!$ in (43) replaces $\mathcal{T}_{\rho,\ell}$ by $\sum_{h=0}^\ell s_h \rho^{-h} \mathcal{T}_{\rho,\ell-h}$ after extracting the corresponding Stirling leading factor. Compactness controls the additional remainder. For a moving, nonconvergent ρ , (44) is an approximation by a moving parameter, not convergence to one fixed distribution. Nothing here treats $\rho \downarrow 0$, $\rho \rightarrow \infty$, or fixed D ; in particular the fixed-dimensional row statement in A328718 is a separate regime.

Remark 9.2 ([write] The rows of A328718, 7 October 2026). The entry A328718 states as a conjecture (Remark 1.1(b)) that its row r , the walks in fixed dimension $D = r$, satisfies

$$A_{N,D}(1) \sim \frac{(2D+1)^{N+D/2}}{2^D(\pi N)^{D/2}} \quad (N \rightarrow \infty, D \geq 0 \text{ fixed}).$$

This is true; it is the local central limit theorem for the lazy walk, and the short proof is as follows. For $D = 0$ both sides equal one. For $D \geq 1$, $A_{N,D}(1) = (2D+1)^N \mathbb{P}(S_N = 0)$ for the walk S_N with independent steps uniform on $\{0, \pm e_1, \dots, \pm e_D\}$, and by Fourier inversion

$$\mathbb{P}(S_N = 0) = (2\pi)^{-D} \int_{[-\pi, \pi]^D} \varphi(\theta)^N d\theta, \quad \varphi(\theta) = \frac{1 + 2 \sum_{i=1}^D \cos \theta_i}{2D+1}.$$

Here φ is real and $|\varphi| \leq 1$; $\varphi(\theta) = 1$ only at $\theta = 0$, and $\varphi = -1$ never, because $1 + 2 \sum_i \cos \theta_i \geq 1 - 2D > -(2D+1)$. So $|\varphi| \leq \gamma < 1$ outside a ball $|\theta| \leq \delta$, which contributes $O(\gamma^N)$. Inside it, $\varphi(\theta) = 1 - |\theta|^2/(2D+1) + O(|\theta|^4)$, so for small δ one has $0 < \varphi(\theta) \leq \exp(-|\theta|^2/(2(2D+1)))$, and after $\theta = x/\sqrt{N}$ dominated convergence gives $N^{D/2} \int_{|\theta| \leq \delta} \varphi^N d\theta \rightarrow \int_{\mathbb{R}^D} e^{-|x|^2/(2D+1)} dx = (\pi(2D+1))^{D/2}$. Hence $\mathbb{P}(S_N = 0) \sim (2\pi)^{-D} (\pi(2D+1)/N)^{D/2} = (2D+1)^{D/2} / (2^D(\pi N)^{D/2})$, which is the displayed formula. Numerically, from exact counts, $N(A_{N,D}(1) 2^D(\pi N)^{D/2} (2D+1)^{-N-D/2} - 1) = -0.1875, -0.3747, -0.5614$ at $N = 400$ for $D = 1, 2, 3$, consistent with a $1/N$ correction. Row 1 is A002426, the central trinomial coefficients. The argument is classical and is a fixed-dimension statement: it neither uses nor extends Theorem 9.1, whose compact-ratio regime excludes fixed D . (Added after the independent check of 7 October 2026: the proof was re-derived step by step and is correct. The $1/N$ term is also explicit. The cubic term of $\log \varphi$ vanishes by symmetry, and its quartic term is $\sum_i \theta_i^4 / (12(2D+1)) - |\theta|^4 / (2(2D+1)^2)$. Its average against the Gaussian weight $e^{-|x|^2/(2D+1)}$ gives $A_{N,D}(1) 2^D(\pi N)^{D/2} (2D+1)^{-N-D/2} = 1 - 3D/(16N) + O(N^{-2})$. That is $-0.1875, -0.375, -0.5625$ for $D = 1, 2, 3$, against the exact-count values $-0.187495, -0.374746, -0.561437$ at $N = 400$ above. Richardson extrapolation from $N = 200$ and 400 gives $-0.1875002, -0.375001$ and -0.562508 . This is the standard Edgeworth computation, recorded here as a numerical corroboration, not as a proved second-order theorem.)

[write] Part II and this section, 7 October 2026. The sentence “Nothing here treats $\rho \downarrow 0$, $\rho \rightarrow \infty$, or fixed D ” describes Part I. Part II treats all three for the counts. Its Theorem 16.2 holds for every D , every N and every nonnegative idle and axis activity, with relative error $O_K(m^{-K-1})$, $m = \lfloor N/2 \rfloor$; Theorem 19.1 makes the leading term explicit for $N/D \rightarrow \infty$, and Theorem 18.1 for $m^2/D = O(1)$. On a compact ρ -interval (a consequence drawn by the write, not stated in Part II), for a real marker $u = \lambda$ in a compact subset of $[0, \infty)$, of $(0, \infty)$ when N is odd, Theorem 16.2 and Theorem 9.1 are two expansions of the same counts with the same relative precision but different leading terms: Part II’s saddle is that of the whole generator $H_\epsilon(\lambda^2 t) F(t)^D$, Part I’s that of F^D , with the idle factor as amplitude. Each leading term is therefore the count times $1 + O_K(1/D)$, so the two agree to relative $O(1/D)$; at $N = D = 40, 41, 100, 101$ and $u = 1$ the write’s evaluation gives Part II’s over Part I’s minus one = $-1.86 \cdot 10^{-3}, -3.65 \cdot 10^{-4}, -7.57 \cdot 10^{-4}, -1.86 \cdot 10^{-4}$. Neither theorem contains the other: Part II has no complex marker, no absolute remainder at the zeros of $E_\epsilon(ru)$ (Remark 4.2) and no zero-step law in this range; Part I has no uniformity in ρ or in anisotropic activities. At fixed $D \geq 1$, Theorem 19.1(i) gives the leading statement of Remark 9.2 by a second route (the parity-resolved saddle and the large-argument Bessel expansion, not Fourier inversion); its error contains $O(1/N)$, so it does not give the $1/N$ term. Subsection 19.1 obtains that term, $-3D/(16N)$, by the same Edgeworth computation as the dated note in Remark 9.2 (not an independent route), with its remainder in outline, and the next coefficient $D(32D^2 - 21D - 8)/1536$, which the write corroborated from exact counts (note at the end of Section 19).

10 Exact checks and reproducibility

The proofs above establish the asymptotic results. Computation checks finite enumeration and coefficient algebra; numerical agreement does not establish a uniform remainder, an effective onset, or priority over prior literature.

10.1 Three finite counting descriptions

An integer convolution avoids cancellation. Put

$$Q_{D,k} = (k!)^2 [t^k] \left(\sum_{h \geq 0} \frac{t^h}{(h!)^2} \right)^D. \quad (45)$$

Then $Q_{0,0} = 1$, $Q_{0,k} = 0$ for $k > 0$, and

$$Q_{D,k} = \sum_{h=0}^k \binom{k}{h}^2 Q_{D-1,k-h}. \quad (46)$$

For $N - j = 2k \geq 0$, the exact number of bridges with j zero steps is

$$\binom{N}{j} \binom{2k}{k} Q_{D,k}. \quad (47)$$

The count is zero at other parities. To count occupied axes independently, put

$$R_{k,m} = (m!)^2 [t^m] \left(\sum_{h \geq 1} \frac{t^h}{(h!)^2} \right)^k.$$

These satisfy $R_{0,0} = 1$ and $R_{k,m} = \sum_{h=1}^m \binom{m}{h}^2 R_{k-1,m-h}$. For $n - j = 2m$,

$$\#\{J_n = j, K_n = k\} = \binom{n}{j} \binom{2m}{m} \binom{n}{k} R_{k,m}. \quad (48)$$

Both recurrences follow by selecting the contribution of one factor in the corresponding generating function.

A different exact calculation uses (6). Write $a(z) = \sum_{k \geq 1} \alpha_k z^{2k}$ and $F(z)^D = \sum_{k \geq 0} \beta_{D,k} z^{2k}$. Then

$$\alpha_k = \frac{4\mathbf{1}_{\{k=1\}} - \sum_{h=1}^{k-1} \alpha_h \alpha_{k-h}}{2k}, \quad (49)$$

$$\beta_{D,0} = 1, \quad \beta_{D,k} = \frac{D}{2k} \sum_{h=1}^k \alpha_h \beta_{D,k-h}. \quad (50)$$

The unmarked count is $N! \sum_{k=0}^{\lfloor N/2 \rfloor} \beta_{D,k} / (N - 2k)!$. Rational arithmetic makes this comparison exact.

10.2 Independent coefficient verification

A raw-derivative route to the cumulants differentiates the second-order Bessel equation instead of iterating (8). Let $g_m = r^m F^{(m)}(r)/F(r)$, with $g_0 = g_1 = 1$. Then

$$\begin{aligned} g_{m+2} = & -(2m+1)g_{m+1} - (m^2 - 4r^2)g_m \\ & + 8mr^2g_{m-1} + 4m(m-1)r^2g_{m-2}, \end{aligned} \quad (51)$$

where terms with negative indices have zero coefficient and are omitted. Converting these factorial moments to raw moments with Stirling numbers, then raw moments to cumulants, checks (9) and higher orders independently.

For the Gaussian algebra, if $\exp(\sum_{h \geq 1} e_h v^h) = \sum_{k \geq 0} P_k v^k$, then

$$P_0 = 1, \quad P_k = \frac{1}{k} \sum_{h=1}^k h e_h P_{k-h}.$$

Take $e_h = \kappa_{h+2}(it)^{h+2}/(h+2)!$, adding ijt to e_1 for monomial amplitude z^j . Integrating each polynomial by Gaussian moments and multiplying the Stirling series gives the $H_\ell(j)$ of (28). Averaging over (25) reproduces the three displayed correction coefficients. This recurrence avoids merely repeating the tuple enumeration in (11).

10.3 Numerical diagnostics

For illustration let $B_n = a_n/(C_\epsilon(dn)^n)$ and $R_2(n) = n^3(B_n - 1 - c_{\epsilon,1}/n - c_{\epsilon,2}/n^2)$. Exact integer counts followed by high-precision evaluation give

n	$R_2(n)$	$n d_{TV}(J_n, J_\epsilon)$	first corrected $n^2 \ell^1/2$
40	-0.011103362959	0.1186222051	0.0468622340
80	-0.011235240759	0.1188802051	0.0471002563
160	-0.011304345034	0.1190101353	0.0472195093
41	-0.070685517647	0.1662254163	0.0765881923
81	-0.070899449934	0.1671477058	0.0766029313
161	-0.071006847771	0.1676175746	0.0766112777

The final column uses the signed approximation $\pi_\epsilon(j)(1 + Q_{\epsilon,1}(j)/n)$, so it is a half ℓ^1 error rather than a distance between two probability laws. The Poisson sums were numerically truncated beyond $n + 100$; they are diagnostics, not outward-rounded total-error certificates. The $R_2(n)$ values approach the two $c_{\epsilon,3}$ in Section 5. Subtracting that term gives stable n^4 -scaled residuals as well.

Separate bivariate counts at $n = 20, 21, 60, 61, 120, 121$ check the occupation mean offsets and characteristic functions. Marked evaluations at the cancellation points in Remark 4.2 test the necessity and correctness of the absolute leading formulation. These are additional numerical checks, not replacements for the complex-uniform proof.

[write] Rechecked, 7 October 2026. From its own exact counts (a rational power of F) the write reproduced every entry of the table above to the printed digits, with the Poisson sums carried to $j = n + 200$. From exact marked counts, $EK_n - np = -0.07093, -0.31307, -0.07109, -0.31237$ at $n = 60, 61, 120, 121$, against the offsets -0.071243 and -0.311664 of (33), with differences halving as n doubles. Observations, not tests of the asymptotic statements.

10.4 Package use and scope

The accompanying ZIP includes this L^AT_EX source, the PDF, standard-library exact verification, optional SymPy/mpmath diagnostics, recorded data, and a deterministic no-shell-escape builder. The core and optional dependencies are segregated. The detailed commands and finite test ranges are listed in `README.md` and `README_CODE.md`; the mathematical verifier can be run with Python’s `-S` flag, without site packages. The builder performs explicit runtime checks also under `-O`, rejects unresolved references and layout defects, and packages a complete SHA-256 inventory. Rebuilding the same sources with the same Python and TeX stack gives byte-identical artifacts; cross-version PDF identity is not promised.

The manifest is an integrity inventory, not a digital signature. The builder does not install dependencies or access the network, and refuses existing output paths. Optional high-precision diagnostics are not required for the exact core or PDF build. No third-party copyrighted book or article PDF is redistributed.

[write] **The shipped layout, 7 October 2026.** In ProveIt the “accompanying ZIP” is not shipped as such: the PDF and the checksum manifest are not shipped, the README is replaced by the report README, and the programs and records are shipped under `code/` and `data/` with prefixed names (Section 1.1); the delivered programs expect the delivered layout, and the report README gives a route that restores it on a copy.

11 Source comparison and limitations

The inspected official OEIS source record already contains the exact Bessel formula and Kotěšovec’s leading equivalent, including both numerical parity constants. Those results are not claimed as new here. The report supplies analytic definitions and a self-contained proof, then develops the stated finite-order, distributional, and inverse refinements. OEIS A328718’s fixed-dimensional row comment is not an unresolved problem being solved by this proportional-dimensional argument.

Gessel–Weinstein–Wilf’s primary article [3] was inspected in full. Its equation (2) gives the Bessel-product walk exponential generating function; its later permutation and Toeplitz-point results concern a different asymptotic framework. Flajolet–Sedgewick’s Theorem VIII.8, full proof, and subsequent vanishing-multiplier remarks [4] were read from the complete author-hosted book. They already provide the generic full-expansion and compact-ratio machinery. Period-two extraction, or the explicit two-arc proof here, handles our even support. We do not infer novelty from writing out that classical method.

A bounded comparison of the ProveIt report corpus at commit `83befe707f2840c2b53e0606701d8a2b28598e47` and of available earlier reports found no matching A328716 treatment in the inspected results. Exact identifier, decimal-constant, Bessel-formula, and lazy-walk searches were used. Nearby reports were actually inspected where relevant: the A039831 two-Fourier-peak maximum-coefficient problem differs from this bridge family, while proposed A047909 and A108242 directions were rejected as already covered. Report 169’s packed-matrix family was also excluded. These comparisons are bounded evidence, not an absence certificate. Earlier-report search missed known duplicates and therefore cannot certify novelty by a negative result.

Searches under Bessel-distribution and conditional-occupancy terminology also found general related literature without identifying this full family-specific package. That search was not exhaustive. Full historical priority for individual probability refinements remains unestablished. The accompanying source audit identifies the inspected sources and these limits. No OEIS record or external repository was edited or submitted as part of this report.

The mathematical limits are also explicit: no growing truncation order, boundary-ratio uniformity, fixed-dimensional asymptotic consequence, total-variation Gaussian limit, or effective numerical inverse threshold is asserted. The weighted discrete law, analytic occupation expansion, and eventual width-one inverse theorem hold precisely in the regimes proved above.

[write] Added 7 October 2026. The write read the A328716 entry itself (Remark 1.1); it agrees with the source’s account of the GitHub record, including both attributions and dates. The “fixed-dimensional row comment” of A328718 is stated there as a conjecture; Remark 9.2 proves it by the classical local limit argument, which is a fixed-dimension argument, so the source’s sentence that it is not “being solved by this proportional-dimensional argument” stays accurate.

12 Further questions within this treatment

Several concrete extensions remain outside the results proved here. These are questions for extending this treatment, not claims that the corresponding problems are historically unsolved.

1. Make the compact saddle bounds effective: produce explicit remainder constants and verified onsets, then combine them with interval evaluations of r , C_ϵ , and the inverse carriers. That would turn the existential brackets of Theorem 8.1 into a certified finite-input inverse calculator.
2. Analyze sparse and dense ratios $N/D \rightarrow 0$ and $N/D \rightarrow \infty$. The compact theorem does not control these limits; the saddle, variance, and relative sizes of the zero-step and coordinate occupancies require separate scaling arguments.
3. Strengthen the occupied-axis weak limit to a lattice local theorem or a joint Edgeworth expansion with the zero-step count. Such a result should explicitly retain the exact support and parity constraints, and specify the range of uniformity and tail norm rather than infer it from the characteristic-function limit alone.
4. Compare the finite-size polynomials in Theorem 6.1 and the occupation corrections systematically with general conditional-occupancy and Bessel-distribution literature. This could clarify alternative formulations and attribution beyond the bounded source search used here.

[write] Added 7 October 2026, under Vladimir’s standing rule of 4 October 2026. The four questions stand as the source states them. The write found no unproved claim of the source outside them and no claim of the source that is wrong; three printed constants are rounded rather than truncated in their last digit (note at the end of Section 3). On question 2: the extreme case of the dense ratio $N/D \rightarrow \infty$ in which D stays fixed is settled for its leading term by Remark 9.2; $N/D \rightarrow \infty$ with $D \rightarrow \infty$, and $N/D \rightarrow 0$, stay open.

[write] Independent check of the write, 7 October 2026. An adversarial check made by the intake after the write (commit `f3fb85cb6`) re-derived the statements marked [write] with its own code, after fetching again A328716 (revision #38), A328718 (#35), A002426 and the A328716 b-file. Remark 9.2, checked the hardest. The reduction to $\mathbb{P}(S_N = 0)$, the Fourier inversion, the two extremal facts ($\varphi = 1$ only at 0; $\varphi > -1$ since $1 + 2 \sum \cos \theta_i \geq 1 - 2D$), the local bound and the Gaussian integral were re-derived, and the conjecture’s row index was confirmed to be the dimension. The exact counts at $N = 400$ (by Miller’s power recurrence for F^D , a route distinct from the source’s two) give the three printed values, and row 1 equals A002426 on all its data terms. A dated note there adds the $1/N$ term $-3D/16$. Remark 1.1. Quotations, revisions and dates confirmed; all 399 b-file terms agree with Miller’s recurrence modulo two primes, and 45 of them ($n \leq 40$, $n = 100, 199, 250, 398$) exactly; Kotěšovec’s d and both c are truncations of d , $C_\pm$. Note of Section 3. At 70 digits, r , b and p end in a rounded digit with the stated truncations and next digits, and d , $C_\pm$, τ^2 are truncations. The note’s scope is clarified there. Notes of Section 10. Every entry of the diagnostics table reproduced, and $\mathbb{E}K_n - np$ computed as $n(1 - A_{n,n-1}(1)/a_n)$ (an axis is unused exactly when the bridge lives in $n - 1$ dimensions): -0.070931 , -0.313069 , -0.071086 , -0.312368 at $n = 60, 61, 120, 121$, as stated.

Remark 8.2. (a)–(e) re-derived against the volume (p0:thm:staircase(4) and p0:rem:parity-instances, p0:prop:factorial-core with $\kappa = 1$, $d_{\text{vol}} = \log d$, the template chart, and the core-reversion form with $\Lambda_{\text{vol}} = 1 + 1/L$). *Provenance.* Archive facts, the pin 83befe707 (3 October 2026), the four neighbouring reports named, and the 72 delivered label numbers and 65 references (aux of a build of the delivered text) confirmed. Apart from the scope of one sentence, no defect was found in the write.

[write] **Part II and these questions, 7 October 2026.** Part II (Section 13) answers part of these questions; the table of Section 13.2 gives the details, and the notes above, which call the two ends of question 2 open, were written before Part II arrived and are kept. *Question 2:* answered for the counts at both ends (Theorem 16.2 for every ratio, Theorem 19.1 and Corollary 19.2 for $N/D \rightarrow \infty$, Theorem 18.1 for $m^2/D = O(1)$), and for the laws of the zero steps and of the repeated and occupied axes in the sparse critical range $m^2/D = O(1)$ (Theorem 18.2). Open: those laws when $N/D \rightarrow \infty$ with $D \rightarrow \infty$, and when $N/D \rightarrow 0$ with $N^2/D \rightarrow \infty$ (Part II's own questions in Sections 21.4 and 21.6 are related). *Question 3:* advanced only in that sparse range, where the occupied axes are not Gaussian: a joint discrete law of zero steps and repeated axes with error $O(1/m)$ in exponentially weighted ℓ^1 , and the occupied-axis deficit $m - K_{\text{occ}}$ in total variation; the local theorem and the joint Edgeworth expansion in the compact range stay open. *Question 1:* advanced in part. Corollary 16.3 gives the explicit two-sided bound $|a_n/L_n - 1| \leq 24/\lfloor n/2 \rfloor$ for every $n \geq 2$, with Part II's leading term L_n ; effective constants for the expansion of Theorem 4.1, interval evaluations and a certified inverse stay open (Part II asks the same for its own expansion in Section 21.3). *Question 4:* untouched.

Part II

Uniform lattice-bridge asymptotics: a Gaussian bound disproved, a sharp dimension threshold, and sparse Poisson laws

A research manuscript of 7 October 2026, archive `uniform-lattice-bridges-research.zip`, delivered as `uniform-lattice-bridges/article.tex` with nine section files (31 pages). Delivered title block “RESEARCH ARTICLE / Uniform Lattice-Bridge Asymptotics / A Gaussian bound disproved, a sharp dimension threshold, and sparse Poisson laws / Prepared for Vladimir Reshetnikov / 7 October 2026”. Printed as delivered after Section 13, which the write added, with [write] notes; the manuscript’s Section k is Section $k + 13$ here, its statements move with their sections, its equation $(k.m)$ is equation $(k+13.m)$, and its Tables 2–4 and Figures 1 and 2 keep their numbers. Its labels carry the prefix `lcw:ulb:`. It uses Part I only as the source of the questions it answers, not as a premise of any proof.

Abstract of this Part (as delivered). We give an exact counterexample to the universal Gaussian return bound conjectured in Felker–Lyons (2003) and stated by Ball–Sterbenz (2005): in dimension 16, the nonlazy return probability after 26 steps is strictly greater than $2(4/(13\pi))^8$. The certificate uses only integer arithmetic and an elementary rational bound for π . A uniform asymptotic theorem explains the failure and proves more: no finite constant can multiply the Gaussian expression into an upper bound valid for every dimension and time.

The main tool is a single expansion for weighted closed walks with steps $0, \pm e_1, \dots, \pm e_D$, uniform in dimension and all nonnegative activities. For length $N = 2m + \epsilon$, exact parity reduction expresses the count through Bessel and hyperbolic functions with negative real zeros. Their tilted coefficients are Poisson–binomial probabilities, with the decisive variance bound $m/2 \leq B \leq m$. We prove expansions to every fixed order, with relative remainder $O_K(m^{-K-1})$, and a leading relative error at most $24/m$.

For isotropic nonlazy and unit-idle walks in the dense range $N/D \rightarrow \infty$, the Gaussian estimate, including the period correction when appropriate, is relatively valid exactly when $D^3/N^2 \rightarrow 0$. At $N/D^{3/2} \rightarrow s > 0$, the missing factor tends to $\exp(1/(48s^2))$. The eventual Gaussian upper-bound thresholds are asymptotic to $D^2/12$ and $D^2/9$, respectively. In the sparse range $m^2/D = O(1)$ with bounded $\lambda\sqrt{m/D}$, repeated axes follow a Poisson law of mean $m^2/(4D)$, asymptotically independent of a parity-conditioned Poisson idle count. We prove error $O(m^{-1})$ in exponentially weighted ℓ^1 , give the first counting correction, and quantify triple occupancy.

The results answer endpoint questions in the repository’s lazy-walk report. Proofs, source comparisons, reproducible checks, and further questions are included. The classical real-zero and Edgeworth methods are credited explicitly; historical priority of the combined conclusions is not asserted.

Mathematical status: proved in the text, with separate computational checks. No Lean formalization or external peer review is claimed.

13 Provenance, scope and notation of Part II

[Added 7 October 2026.] Part II was added in a second write of this report, after Part I had been written (`f3fb85cb6`) and independently checked (`1ba9be08f`). It is the manuscript *Uniform Lattice-Bridge Asymptotics*, which starts from Part I’s Question 2 (Section 12) and answers it for the counts. Everything of the manuscript is printed in this Part except its title page (recorded in the source block above and below), its table of contents and its reference list (merged into this

report’s bibliography): its abstract and status line above, and its eight sections as Sections 14–21, in the manuscript’s order. Its letters are kept (Section 13.5). This section and the notes marked [write] are the write’s; all other text of Sections 14–21 is the manuscript’s.

13.1 The manuscript

The archive `uniform-lattice-bridges-research.zip` (838,687 bytes, SHA-256 `f736b7f99454...3eaff7bb6`; 24 files in one wrapper directory `uniform-lattice-bridges/`, 1,057,955 bytes unpacked; no checksum manifest) arrived in commit `fe7165a3c` (7 October 2026, 10:58). Its 12 supporting files were placed in this directory by commit `3cf0a2758` (7 October 2026, 13:16), which also retired the archive, with the prefix `02-bridges-`, the Part number; all 12 are byte-identical to the delivery (confirmed at the write). The programs are `code/02-bridges-*.py`; the verification record, the generated table, `requirements.txt` and `source_provenance.json` are under `data/`; the two figures (PDF, included here, and PNG previews) under `figures/`. Not shipped, and retrievable from the arrival commit: the manuscript sources and its PDF, and the delivery `README.txt`.

The manuscript is `article.tex` with `sections/01_context.tex` to `sections/08_outlook.tex` and `sections/references.tex` (2,139 lines, 88,397 bytes; a 31-page PDF, which a build of the delivered text reproduces), dated 7 October 2026. Its driver inputs the section files in the order 01, 06, 02, 03, 04, 05, 07, 08, so the counterexample file is its Section 2. Its title page reads “Prepared for Vladimir Reshetnikov”, its PDF author field and its README “Research manuscript prepared for Vladimir Reshetnikov”; it names no person or tool as author and carries no AI-assistance line (none is supplied here). Its pins (Section 14, `data/02-bridges-source_provenance.json` and the bibliography entries [5, 6]) are ProveIt commit `136e707021c9df3a1b6af904ddad045ae38b7c23` (7 October 2026, 17:07:16 UTC), which is later than Part I’s write and its independent check, so the manuscript read Part I as written, and commit `adc7f1241b42` of Vladimir’s `math` repository, which is outside this repository and was not checked. Part I’s `article.tex` is the same at the pin as before this write.

How the manuscript was merged. Its 76 labels carry the prefix `lcw:ulb::` `ulb:X` and unprefix `X` became `lcw:ulb:X`, and the sparse section’s `newsp:X` became `lcw:ulb:sp:X` (the two sections both use `eq:exact`); its 75 references to them (`41 \eqref`, `34 \ref`) were updated. The three table labels live in the shipped file `data/02-bridges-numerical_tables.tex`, which is input unchanged through a local wrapper that adds the prefix. The table file and the two figures are input under their prefixed names. Its citation keys are new entries of the merged bibliography, marked [Part II], except `gww`, which Part I cites too and which is printed once with Part II’s link and note added; its key `lcw` is this report at the pin, that is, Part I. Its macros (`\Prob`, `\Dop`, `\ep`, `\R`, `\Z`, `\sech`, `\csch`, `\repo`, `\sourcepin`, and the unused `\N`, `\Pois`, `\TV`, `\abs`) and the packages `graphicx` and `longtable` were added to the preamble; its `\E` and `\Var` are Part I’s. The title page and table of contents are not printed. No other text was changed; the survey table of Section 14 is set ragged right (layout only), and the unlabelled sparse section received the label `lcw:ulb:sp:sec:sparse`. The numbering, against a build of the delivered text: its Sections 1–8 are Sections 14–21 here, its Theorem 2.1 is Theorem 15.1, Theorem 3.2 is Theorem 16.2, Corollary 3.3 is Corollary 16.3, Theorems 5.1 and 5.2 are Theorems 18.1 and 18.2, Theorem 6.1 is Theorem 19.1, every statement, subsection and equation $k.m$ is $(k+13).m$, and Tables 2–4 (the uncaptioned table of Section 14 takes number 1, as delivered) and Figures 1 and 2 are unchanged (all 76 labels compared). The manuscript cites by number only Part I’s Sections 9 and 12, which keep their numbers, and outside sources.

What the manuscript says about the repository. Its account of Part I (the compact-ratio Theorem 9.1, Question 2 of Section 12, and the fixed- D Remark 9.2 with the note that the dense

case with growing dimension stays open) is correct. Its table of the other threads it surveyed is context, not a premise of any proof; one entry was true at its pin and is stale now (dated note at the end of Section 14).

13.2 What Part II answers in Part I

Part I	After Part II
Question 2 (the ratios $N/D \rightarrow 0$ and $N/D \rightarrow \infty$)	Answered for the counts at both ends: Theorem 16.2 for every D , N and nonnegative activity; Theorem 19.1 and Corollaries 19.2–19.4 for $N/D \rightarrow \infty$, with the explicit $-3D/(16N) + D^3/(48N^2)$; Theorem 18.1 for $m^2/D = O(1)$, with the first correction. Answered for the laws in the sparse critical range $m^2/D = O(1)$: zero steps, repeated axes and the occupied-axis deficit (Theorem 18.2). Open: the laws for $N/D \rightarrow \infty$ with $D \rightarrow \infty$, and for $N/D \rightarrow 0$ with $N^2/D \rightarrow \infty$.
Question 3 (local theorem or joint Edgeworth expansion for the occupied axes)	Advanced only in the sparse critical range , where the occupied axes are not Gaussian: the joint law of zero steps and repeated axes with error $O(1/m)$ in exponentially weighted ℓ^1 , and the deficit $m - K_{\text{occ}}$ in total variation. The compact-range question stays open.
Question 1 (effective constants, certified inverse)	Advanced in part: the explicit bound $ W_N/L_N - 1 \leq 24/m$ (Corollary 16.3), hence $ a_n/L_n - 1 \leq 24/\lfloor n/2 \rfloor$ for every $n \geq 2$. Effective constants for Theorem 4.1, interval evaluations and a certified inverse stay open.
Question 4 (comparison with the literature)	Untouched.
Remark 9.2 (rows of A328718)	Its leading statement follows from Theorem 19.1(i) at fixed $D \geq 1$ by a second route; Subsection 19.1 computes its $1/N$ term by the same Edgeworth computation as its dated note, and the next term.
Theorem 9.1	Not contained in Theorem 16.2: on its common range (real marker, compact ratio) the two give expansions of the same counts with leading terms that agree to relative $O(1/D)$ (note at the end of Section 9).

Dated notes at the ends of Sections 9 and 12 say the same; the earlier notes there, which call the two ends of Question 2 open, are kept as written.

What Part II proves again. Its exact identity (Proposition 16.1) contains Part I’s count (2) (at $u = \lambda$, after splitting the parity of the idle steps), and its fixed- D subsection repeats Part I’s Edgeworth $1/N$ term. *What is new to the report:* everything else, in particular the counterexample, which is new to the repository (before the placement no file of the repository mentioned Felker or Sterbenz).

13.3 What the write read and checked

At placement (the intake dossier, 7 October 2026): the three shipped programs rerun on a copy (`verify_lyons.py` and `verify_endpoints.py`, standard library; `reproduce.py` with `mpmath` 1.3.0); the record equal except for the Python version, the table file equal apart from line ends; the counterexample confirmed exactly by an earlier batch’s own count; no claim found wrong.

At the write (7 October 2026; Python 3.14.4, `mpmath` 1.3.0, `SymPy` 1.14.0; own code, kept with the intake records and not shipped). Every proof of Sections 15–19 was read line by line, with no error found; rechecked by hand in particular: the three formulas of Lemma 16.4 and the Bernoulli bound $B \leq m$; the integral representation of F_ν ; every constant of Proposition 17.3;

the transfer inequalities and the expansion (18.12), its triple-use coefficient and the corrections C_ϵ and their unweighted forms; the conversion of the parity saddle in the proof of Theorem 19.1 and its two corollaries on signs and thresholds. With SymPy it rederived (19.6), (19.8), the prefactor expansion, the totals $-3/(16\rho)$ and $D/(48\rho^2)$ and the logarithm of the fixed- D expansion. Exactly, it recounted the counterexample by a third route and decided it in interval arithmetic, and scanned the conjecture for $d \leq 40$ (note at the end of Section 15); it reproduced every entry of Table 4 and checked further sparse cases and $\mathbb{E}U$ (Section 18); it computed R/G at $D = 100$ from exact counts and recomputed all of Table 3 by its own saddle code, and the fixed- D second coefficient from exact counts (Section 19); it checked the envelope (16.11) on an anisotropic example (Section 16). It reran the three shipped programs on a copy (Section 20). It read the author-hosted Ball–Sterbenz preprint and the Felker–Lyons version of 6 November 2003 at the cited places (note at the end of Section 15). *Not read by the write*: the journal versions of both papers, the Felker–Lyons erratum, Bender, Pitman, Dolgopyat–Hafouta, the DLMF sections and the `math` repository; the manuscript’s accounts of them are reported, not confirmed.

13.4 What is not claimed

From the manuscript: no historical priority for the counterexample or the combined conclusions, and no minimum dimension or length (“No claim that this is the smallest possible dimension or the shortest possible counterexample is needed”; its README: the literature audit “does not establish historical priority or a minimum dimension”); the final Ball–Sterbenz journal PDF not inspected; the real-zero, Bernoulli and Edgeworth methods classical; Poincaré expansions at fixed order, with existential constants C_K ; the dense large-dimension table entries are saddle evaluations, not enumerations; no interval arithmetic (the $24/m$ envelope becomes a numerical interval only with outward-rounded evaluation of the saddle); the sign corollary asserts no uniqueness of crossings; the effective dimension of Section 21.5 is “a proposed organizing parameter”; the review of the two repositories was bounded and infers no formal scope; nothing formalized, no external review. The write adds: its checks are exact finite computations, floating evaluations or hand computations; its scan of the conjecture is finite and settles no minimum; nothing was submitted anywhere.

13.5 Reading conventions

Part II keeps its letters. Where they clash with Part I, read them as follows (the write’s own text in Part II uses Part II’s letters).

Symbol	Part II	Part I
ϵ	the parity remainder in $N = 2m + \epsilon$, $\epsilon \in \{0, 1\}$	the sign $(-1)^n$; Part I’s +1 is Part II’s 0
m, N	$N = 2m + \epsilon$ the length, m the number of pairs	N the length, $n = N = D$ on the diagonal
F	$F(t) = \sum t^k / (k!)^2 = I_0(2\sqrt{t})$	$F(z) = I_0(2z)$, Part II’s F at $t = z^2$
$\mathcal{D}$	$t d/dt$	$\mathcal{D} = z d/dz$; under $t = z^2$, $t d/dt = \frac{1}{2}z d/dz$
r, κ_j, B	the saddle of G_ϵ in t (mean m) and its cumulants and variance; B also a bound for β (Theorem 18.1) and $B(z) = z d/dz \log I_0(2z)$ (Section 19)	r the saddle of F in z (mean 1), $\kappa_h, b = 4r^2 - 1$; $B_w(z)$
E_s	the saddle corrections (16.7), Part I’s $\mathcal{T}_s$ of (11) applied to the constant 1 with (B, κ_j) for (b, κ_h)	$E_\epsilon(t) = e^t + \epsilon e^{-t}$, $E_{\epsilon,u}$

Symbol	Part II	Part I
H_0, H_1, h	$\cosh \sqrt{t}, \sinh \sqrt{t}/\sqrt{t}; h = x _1, h = B^{-1/2}$ (Section 17)	$H(v, t), H(s), H_\ell(j); h$ a cumulant order, $h(w)$
λ, J	the idle activity (Part I's marker u); the number of idle steps (Part I's J_n); $J = \sum_j j\nu_j$ in (16.7)	$u; J_n, J_\epsilon$
K_{occ}, R, U, K	occupied (Part I's K_n), repeated and triple-use axes; K a fixed order, a bound for θ , K_i axis counts	$K_n; R_2(n), R_{k,m}; U_M; K$ a compact interval
θ, β	$\theta = m^2/D, \beta = \lambda\sqrt{m/D}$; θ also an angle	θ an angle
C, c	$C_K; C_\epsilon(\theta, \beta)$ the sparse correction (18.3); $C_{m,d}$ multinomial sums; $c_i = a_i b_i$; $c = \beta^2$; $c = 1 + 1/(2D)$	$C_\epsilon = E_\epsilon(r)/\sqrt{b}$; $c_{\epsilon,\ell}$
ρ, q	$\rho = N/D$ (as in Part I), $q = D/N$	$\rho = N/D$; $q = 1/F(r)$
$R_{N,D}, G_{N,D}$	the lazy return probability and its Gaussian prediction; R^0, G^0 nonlazy	$R_2(n), R_{k,m}$
Q	$Q_{n,D} = (n!)^2[t^n]F(t)^D$, Part I's $Q_{D,n}$ (indices swapped); $Q_\epsilon(z) = 1 + (-1)^\epsilon e^{-2z}$; $Q \sim \mathcal{P}_{\epsilon,\beta}$; $Q(t, x)$	$Q_{D,k}, Q_{\epsilon,\ell}$
T, L	T_m the transfer; $T_0(D), T_1(D)$ thresholds; $L_N, L_{N,x}$ leading terms; L an order	$\mathcal{T}_\ell, T(n, k); L_M$ a lower bracket
$\mathcal{P}_{\epsilon,\beta}$	Poisson(β) conditioned on parity ϵ	π_ϵ , Poisson(r) conditioned on the sign ϵ
a, b, d, s, p	activities a_i, b_i ; $a = 2D + 1, a = 15/32$, $a \in \{1/4, 3/16\}$; d the dimension (Section 15); s a limit of $N/D^{3/2}$ or N/D^2 , an index; p_j Bernoulli parameters	$a_n; b; d = F(r)/(er)$; s_j Stirling coefficients; $p = 1 - q$
X, x, ν	X_j Bernoulli variables; x an endpoint or a scaled angle; $\nu_i = x_i , \nu_j$ multiplicities	$X_r, X; x = z^2; \nu(X)$

14 Repository review and the research question

The starting request was to review recent results in `VladimirReshetnikov/ProveIt` and `VladimirReshetnikov/math`, select a promising direction, and develop substantial new mathematics. The review was pinned to the `ProveIt` commit `136e707021c9` and the `math` import `adc7f1241b42` [5, 6]. Pinning matters: the Ramsey corpus had already absorbed several rounds of improvements on the day of inspection. A question in an earlier report may have been answered in a later report without the older question being deleted.

14.1 Results reviewed and their precise scope

The following table records the mathematical reasons for the selection. Statements about formalization are reports of the inspected repository records; no repository-wide Lean rebuild was performed for this article. In particular, a manuscript, its formalization scope document, and a quantitative consequence are treated as separate objects.

Direction	Inspected progress	Consequence for selection
Gowers–Szemerédi formalization in ProveIt	Recent commits record weighted slab product estimates, compressed interpolation, and improved explicit four- and five-term bounds relative to the source parameters. The ledger distinguishes 113 exact companions from seven unresolved catalogue entries. One packaged Lemma 16.10 interface has a counterexample, so an interface repair and a proof are different tasks.	A further global bound would require resolving substantial structural obligations, not merely replacing a local constant.
Boolean phase integration, sources 65 and 69	The two reports obtain the all-degree nonintegrable energy extremum $1 - (d + 1)/2^d$, with a tensor equality classification. Their explicit remaining questions include odd characteristic and maximizing phases.	Reproving the all-degree value would duplicate the newest intake. The remaining analytic questions are serious but were not resolved in this investigation.
math , family 159	The catalogue describes quasipolynomial arithmetic-progression bounds. The scope document identifies the formalized comparator as the reciprocal-sum consequence, and explicitly separates the quantitative extremal estimate [7].	Formal scope cannot be inferred from the family title. This result is not an input to the present proof.
math , families 241 and 242	The scope files describe full Turing-degree rigidity and single-fold Diophantine representation statements, while excluding specified representation or undecidability corollaries [8, 9]. Selected manuscript arguments and the catalogue were inspected.	These suggest substantial future work, but a sequel would require specialized foundations. The present article does not certify or assume these claims.
Lazy closed walks, A328716/A328718 report	The report proves fixed-order expansions in a compact positive ratio range. A later note supplies the classical fixed-dimension leading term. The sparse limit and the dense limit with growing dimension remain explicit questions [10].	There is a precise, self-contained gap with a promising structural solution: remove the period-two obstruction before applying a uniform coefficient theorem.

The **math** README describes 722 manuscripts in 372 families at different verification stages [6]. The catalogue review found no indicated Bessel/Poisson–binomial lattice-return treatment overlapping the theorem below. This is a bounded comparison with the material inspected, not a certificate of absence from every manuscript or from the literature.

14.2 The exact gap in the predecessor

The immediate predecessor is *Lazy Closed Walks in Growing Dimension: All fixed orders, parity conditioned Poisson laws, and occupied coordinate fluctuations for OEIS A328716*, archived under `a328716-lazy-closed-walks` [10]. Its proportional dimension theorem uses the parameter $\rho = N/D$ in compact subsets of $(0, \infty)$. Section 12 asks for both $\rho \rightarrow 0$ and $\rho \rightarrow \infty$. The integrated edition clarifies that the growing-dimensional dense case remains open after its fixed- D remark. These are questions internal to that report; they are not presented here as long-standing named conjectures in the published literature. The dense analysis also led to the separate refutation in Section 15 of a universal bound appearing in [11, 12].

The new results settle the counting part of both endpoint questions and also give a sharp criterion for the Gaussian simplification in the dense case. They settle the sparse joint lattice-law question in the critical collision range. They give an explicit effective leading error, while a complete high-order interval-arithmetic evaluator remains a further task.

New result	Main conclusion
Theorem 15.1 and Corollary 19.2	An exact counterexample to the published universal Gaussian conjecture, and failure of every dimension-independent constant replacement.
Theorem 16.2	Relative expansions of every fixed order for all dimensions and nonnegative activities; no ratio restriction.
Corollary 16.3	A leading saddle approximation with relative error at most $24/m$.
Theorems 18.1 and 18.2	A uniform sparse correction and a joint Poisson law in a strong discrete norm.
Theorem 19.1	The Gaussian estimate is relatively correct precisely for $D^3/N^2 \rightarrow 0$ in the dense range; its critical missing factor and quadratic eventual-bound thresholds are explicit.

14.3 Established methods and the novelty boundary

The Bessel generating function belongs to classical lattice-walk enumeration; Gessel, Weinstein, and Wilf provide relevant background [3]. Negative real zeros and independent Bernoulli representations have a long history in asymptotic enumeration, including Bender and Pitman [14, 15]. General local Edgeworth theory is considerably broader than the special lemma proved here; see Dolgopyat and Hafouta [16]. We give the short complete proof needed for the present uniformity rather than claiming these tools as new.

The new work is the combination of an exact parity reduction, a uniform variance lower bound, and the resulting endpoint analysis. The dense threshold needs a uniform remainder: a fixed-dimension Edgeworth coefficient alone does not prove it. The sparse statement gives a joint law with a quantitative error, not just a pointwise generating-function limit. Historical priority for the full collection of conclusions is not established by this targeted review. Each comparison with the predecessor is stated at the level actually proved.

[write] **The review at the pin, 7 October 2026.** The rows of the table above were checked against the repository at the pin 136e70702 only as far as this repository goes. The lattice-walk row and the next subsection describe Part I correctly (its Section 9 is Section 9, its Section 12 is Section 12, and the “integrated edition” is Part I’s write with its dated notes). The Gowers–Szemerédi ledger (`Combinatorics/Ramsey/gowers-proof-status.json`) records 113 exact companions and 7 open statements at the

pin, as stated; it records 114 and 6 at this write, since the repair of the Lemma 16.10 interface the row mentions (commit 456b6e3ad, after the pin). The Boolean row refers to the reports numbered 65 and 69 in `Combinatorics/Ramsey/Research/GowersSzemerédi/local-quantitative-refinements/`, which state the value $1 - (d+1)/2^d$; they were not checked further. The `math` rows were not checked. None of these rows is a premise of any proof in Part II.

15 A finite counterexample to the Gaussian return bound

Ball and Sterbenz, in *Explicit Bounds for the Return Probability of Simple Random Walks*, Journal of Theoretical Probability 18 (2005), 317–326, state the following conjecture of Russ Lyons in equation (1.3) of the available preprint dated 19 January 2004 [12]: for the nonlazy simple random walk on $\mathbb{Z}^d$,

$$p_{2n}^{(d)}(0,0) \leq 2 \left(\frac{d}{4\pi n} \right)^{d/2} \quad (d \geq 1, n \geq 1). \quad (15.1)$$

The same universal inequality is conjectured in Felker and Lyons [11], equation (2.3). The leading factor 2 is the lattice-period correction. Ball and Sterbenz’s Proposition 1 and Remark 1 establish a sufficient large-time range of order d^3 , and they report checks through dimension six. The universal inequality as stated is false.

Theorem 15.1 (An exact finite counterexample). *For the nonlazy simple random walk in dimension $d = 16$, after 26 steps,*

$$p_{26}^{(16)}(0,0) = \frac{5630493790109507827003978475}{2^{118}} > 2 \left(\frac{4}{13\pi} \right)^8.$$

Indeed, the ratio of the left-hand side to the right-hand side is strictly greater than 4001/4000.

Proof. Define the integers

$$C_{m,d} = \sum_{k_1 + \dots + k_d = m} \binom{m}{k_1, \dots, k_d}^2.$$

Splitting off the last coordinate gives the finite exact recurrence

$$C_{m,1} = 1, \quad C_{m,d+1} = \sum_{k=0}^m \binom{m}{k}^2 C_{m-k,d}.$$

A closed walk uses the positive and negative direction of each axis equally often. Its count is therefore $\binom{2m}{m} C_{m,d}$, and hence

$$p_{2m}^{(d)}(0,0) = \frac{\binom{2m}{m} C_{m,d}}{(2d)^{2m}}.$$

At $(m,d) = (13,16)$ the recurrence gives

$$C_{13,16} = 2217420395389549070189056,$$

and thus exactly

$$\binom{26}{13} C_{13,16} = 23062502564288544059408295833600$$

closed walks. Division by 32^{26} gives the displayed probability. The accompanying verification script performs this recurrence and independently obtains the same count by repeated squaring of the degree-13 truncation of $\sum_{k \geq 0} t^k / (k!)^2$.

No floating-point comparison with π is required. The classical rational lower bound $\pi > 333/106$ implies

$$2 \left(\frac{4}{13\pi} \right)^8 < 2 \left(\frac{424}{4329} \right)^8.$$

For a self-contained certificate of this lower bound, Machin's identity $\pi = 16 \arctan(1/5) - 4 \arctan(1/239)$ and the alternating-series bounds give

$$\pi > 16 \left(\frac{1}{5} - \frac{1}{3 \cdot 5^3} + \frac{1}{5 \cdot 5^5} - \frac{1}{7 \cdot 5^7} \right) - \frac{4}{239} > \frac{333}{106};$$

the final comparison is rational arithmetic. Machin's identity itself follows from the tangent addition formula and the fact that the relevant angle lies in $(0, \pi/2)$.

Writing $A = 5630493790109507827003978475$, the needed comparison reduces to the positive integer

$$A 4329^8 - 2^{143} 53^8 = 241148999863792122048803143474490963808862899634921387 > 0.$$

The stronger ratio bound follows from the exact comparison

$$4000A 4329^8 > 4001 \cdot 2^{143} 53^8.$$

Every displayed integer and rational comparison is checked by the standard-library-only script `verify_lyons.py`. $\square$

The numerical values, included only to show the size of the effect, are

$$p_{26}^{(16)}(0, 0) \simeq 1.6943650925091064 \times 10^{-8}, \quad 2 \left(\frac{4}{13\pi} \right)^8 \simeq 1.6934178227188187 \times 10^{-8}.$$

Their ratio is approximately 1.0005593833828717. The exact certificate, rather than these decimal approximations, proves the counterexample. No claim that this is the smallest possible dimension or the shortest possible counterexample is needed.

Source locator. Primary paper: <https://doi.org/10.1007/s10959-003-2606-7>. The author-hosted full text is https://mathweb.ucsd.edu/~bdriver/DRIVER/Miscellaneous_Files/sterbenz6.pdf; see equation (1.3) on page 2, Proposition 1 and Remark 1 on pages 3–4.

15.1 What this finite calculation settles

This disproves the universal inequality (15.1). It is consistent with the eventual fixed-dimension upper bound proved in [12]. More substantially, Corollary 19.2 below proves that the ratio to the period-corrected Gaussian expression is unbounded along even lengths $N \sim D^{4/3}$. Thus replacing the factor 2 by any other fixed finite constant cannot repair the universal statement.

For the historical comparison, the author-hosted Felker–Lyons version of 6 November 2003, the Ball–Sterbenz paper, and the linked Felker–Lyons erratum of 7 August 2012 were inspected. That erratum concerns body-centered lattice formulas and numerical bounds [13]. A targeted search of later references located no prior refutation of (15.1); this search is not an exhaustive priority determination. The mathematical claim here is the explicit counterexample and the proved asymptotic failure of every constant bound.

[write] **Checked, 7 October 2026.** *The certificate.* With its own code the write counted the closed walks of length 26 in $\mathbb{Z}^{16}$ by a third route, with no power series and no recurrence over the dimension: for each of the 101 partitions of 13 it counted the assignments of the parts to axes and summed the multinomial numbers $26! / \prod_i k_i!^{i^2}$. It obtained 23,062,502,564,288,544,059,408,295,833,600 = $\binom{26}{13} C_{13,16}$ with $C_{13,16}$ as printed, A as printed (the count divided by 2^{12} , an odd integer), the printed positive integer, and $4000A \cdot 4329^8 > 4001 \cdot 2^{143} 53^8$. The four Machin terms give $\pi > 3.1415916745 > 333/106$; independently, the partial sums of the positive series $\pi = 6 \arcsin \frac{1}{2}$ give $\pi > 3.1415926521$. In mpmath interval arithmetic (160 bits) the lower end of the ratio's interval is 1.000559383382871717665...; with π replaced by 333/106 the certificate keeps the margin 1.000347..., so its bound 4001/4000 holds. The three displayed decimals agree with $1.694365092509106442... \times 10^{-8}$, $1.6934178227188187305... \times 10^{-8}$ and $1.0005593833828717176...$. The shipped code/02-bridges-verify_lyons.py, rerun on a copy, prints the same count and difference.

The sources. The write read the author-hosted Ball–Sterbenz preprint cited above (its date line is 19 January 2004; the PDF was produced in 2005 from walk_bound_jtp.dvi). Its display (1.3), on page 2, is (15.1), stated for $d \geq 1$, $n \geq 0$ (above: $n \geq 1$; at $n = 0$ the right side is undefined, so nothing changes), as a conjecture of Russ Lyons. Proposition 1 (page 3, for $2 \leq n$ and $d \leq n$) and Remark 1 (pages 3–4: $n \geq \frac{7}{4}d^2(d+2) + 1$ suffices, so the range is of order d^3) and the computer checks for $d = 3, 4, 5, 6$ (page 4; $d = 1, 2$ being easy, page 2) are as described. In the Felker–Lyons version of 6 November 2003, page 4, the authors say that the inequality appears to be true for all d and k , and then display (2.3) only as known to hold for all k when $1 \leq d \leq 6$ and for all large k (“if not all k ”) when $d \geq 7$. So the universal inequality is their stated belief; the display, as qualified, is not contradicted. Their rigorous bounds for $h_3, \dots, h_6$ use (2.3) only for $3 \leq d \leq 6$, which the counterexample does not touch (compare Section 21.9). The journal versions and the erratum were not read.

A finite scan (computed by the write). Exact counts for $1 \leq d \leq 40$ and $1 \leq n \leq 200$ (walks of length at most 400), and for $d \leq 16$ up to $n = 500$, each comparison with (15.1) decided in interval arithmetic: for $d = 16$ the only violation is $n = 13$, the counterexample above; for each $17 \leq d \leq 40$ the violations form one run of consecutive n ($n = 12, \dots, 16$ for $d = 17$; $n = 19, \dots, 86$ for $d = 40$, with largest ratio 1.3225... at $n = 28$); for $d \leq 15$ there is none, and the largest ratio there, below one, is at the end of the range, $n = 500$. This is an observation in a finite range: for $d \leq 15$ it stops far below Ball–Sterbenz’s guaranteed range (from $n = 6695$ at $d = 15$), so it does not show that 16 is the least dimension, which the manuscript does not claim either.

16 An expansion uniform in dimension and all activities

16.1 Weighted bridges and exact parity reduction

Fix a nonnegative integer D . Give the step $+e_i$ activity $a_i \geq 0$, the step $-e_i$ activity $b_i \geq 0$, and the step 0 activity $\lambda \geq 0$. The weight of a walk is the product of its step activities. Let $W_N(\mathbf{a}, \mathbf{b}, \lambda)$ be the total weight of length- N walks starting and ending at the origin. Only $c_i = a_i b_i$ matters for a closed walk, so we also write $W_N(\mathbf{c}, \lambda)$. The isotropic notation is $A_{N,D}(\lambda) = W_N((1, \dots, 1), \lambda)$.

These weights include ordinary counts, colored idle steps when λ is an integer, and transition probabilities. If $\lambda + \sum_i (a_i + b_i) = 1$, then W_N is already the return probability. For any positive total activity, division by that sum to the power N normalizes the step activities. No symmetry assumption $a_i = b_i$ is required.

Define the entire functions, by their power series,

$$F(t) = \sum_{k \geq 0} \frac{t^k}{(k!)^2}, \quad H_0(t) = \sum_{h \geq 0} \frac{t^h}{(2h)!}, \quad H_1(t) = \sum_{h \geq 0} \frac{t^h}{(2h+1)!}. \quad (16.1)$$

For positive arguments these are $I_0(2\sqrt{t})$, $\cosh \sqrt{t}$, and $\sinh \sqrt{t}/\sqrt{t}$. Writing them by series avoids

any square-root branch convention. For $N = 2m + \epsilon$, $\epsilon \in \{0, 1\}$, put

$$G_\epsilon(t) = H_\epsilon(\lambda^2 t) \prod_{i=1}^D F(c_i t). \quad (16.2)$$

Proposition 16.1 (Exact coefficient identity). *For every nonnegative choice of activities,*

$$W_N(\mathbf{c}, \lambda) = N! \lambda^\epsilon [t^m] G_\epsilon(t). \quad (16.3)$$

Proof. If axis i is used k_i times in each direction and there are j idle steps, then $j + 2 \sum_i k_i = N$. The number of orders of these steps is $N! / (j! \prod_i (k_i!)^2)$ and their common weight is $\lambda^j \prod_i c_i^{k_i}$. Set $j = 2h + \epsilon$ and sum. Equivalently, the exponential generating function before splitting parity is $e^{\lambda z} \prod_i I_0(2\sqrt{c_i} z)$. $\square$

If $\epsilon = 1$ and $\lambda = 0$, the original count is exactly zero. For all other cases with at least one nonconstant factor in G_ϵ , the following gives a relative asymptotic. The normalized coefficient $[t^m] G_1(t)$ itself continues smoothly to $\lambda = 0$ and also has the stated expansion. If all c_i and λ vanish, then $G_\epsilon \equiv 1$ and its positive-degree coefficients vanish; no positive saddle exists in this degenerate case.

16.2 The saddle and the main theorem

Write $\mathcal{D} = t d/dt$. For a nonconstant generator G_ϵ , let $r > 0$ be the unique solution of

$$(\mathcal{D} \log G_\epsilon)(r) = m. \quad (16.4)$$

Existence and uniqueness, for $m > 0$, follow in the proof below. Define the cumulants and leading approximation by

$$\kappa_j = (\mathcal{D}^j \log G_\epsilon)(r), \quad B = \kappa_2, \quad L_N = \frac{N! \lambda^\epsilon G_\epsilon(r)}{r^m \sqrt{2\pi B}}. \quad (16.5)$$

When discussing the normalized odd coefficient at zero, omit the factor λ^ϵ in both the exact count and L_N .

Theorem 16.2 (Uniform expansion in all parameters). *For every fixed integer $K \geq 0$ there are constants C_K, m_K , depending only on K , such that for every $m \geq m_K$, every dimension D , every parity, and every allowed nonnegative choice of activities with nonzero count,*

$$\frac{W_N(\mathbf{c}, \lambda)}{L_N} = \sum_{s=0}^K E_s + \mathcal{R}_K, \quad |\mathcal{R}_K| \leq C_K m^{-K-1}. \quad (16.6)$$

Here $E_0 = 1$. For $s \geq 1$, sum over nonnegative integers $\nu_3, \dots, \nu_{2s+2}$ satisfying $\sum_{j=3}^{2s+2} (j-2)\nu_j = 2s$, and set $J = \sum_j j\nu_j$. Then

$$E_s = \sum_{\sum (j-2)\nu_j = 2s} \frac{(-1)^{J/2} (J-1)!!}{B^{J/2}} \prod_{j=3}^{2s+2} \frac{\kappa_j^{\nu_j}}{\nu_j! (j!)^{\nu_j}}. \quad (16.7)$$

The integer J is even. Each $E_s = O_s(m^{-s})$ uniformly in all the same parameters, and

$$\frac{m}{2} \leq B \leq m. \quad (16.8)$$

The dimension and the activities may depend arbitrarily on m .

In particular, the first two corrections are

$$E_1 = \frac{\kappa_4}{8B^2} - \frac{5\kappa_3^2}{24B^3}, \quad (16.9)$$

$$E_2 = -\frac{\kappa_6}{48B^3} + \frac{7\kappa_3\kappa_5}{48B^4} + \frac{35\kappa_4^2}{384B^4} - \frac{35\kappa_3^2\kappa_4}{64B^5} + \frac{385\kappa_3^4}{1152B^6}. \quad (16.10)$$

The expression is a fixed-order Poincaré expansion. There is no assertion that the infinite sum of its corrections converges, or that one may let K grow without additional estimates.

Corollary 16.3 (Explicit leading envelope). *For every $m > 0$ in the nonzero cases,*

$$\left| \frac{W_N}{L_N} - 1 \right| \leq \frac{12}{B} \leq \frac{24}{m}. \quad (16.11)$$

Consequently, for $m > 24$,

$$L_N(1 - 24/m) \leq W_N \leq L_N(1 + 24/m).$$

The same inequalities apply to the normalized odd coefficient at $\lambda = 0$.

The envelope is deliberately conservative. Its value is that its constant and its parameter range are explicit, with no hidden dependence on dimension or anisotropy. Evaluating $r, G_\epsilon(r), B$ by outward-rounded arithmetic would turn it into a numerical interval. The supplied floating-point examples illustrate the formula; they do not by themselves implement outward rounding.

16.3 Negative zeros and the variance sandwich

Let $j_{0,k}$ denote the positive zeros of the ordinary Bessel function J_0 . The classical product identities [17, 18] give

$$F(t) = \prod_{k \geq 1} \left(1 + \frac{4t}{j_{0,k}^2} \right), \quad (16.12)$$

$$H_0(t) = \prod_{k \geq 0} \left(1 + \frac{t}{\pi^2(k + \frac{1}{2})^2} \right), \quad H_1(t) = \prod_{k \geq 1} \left(1 + \frac{t}{\pi^2 k^2} \right). \quad (16.13)$$

Their reciprocal zero moduli are summable, so the products and their logarithmic derivatives converge locally away from the zeros. Scaling the argument by any positive activity preserves this property; a zero activity contributes the constant factor one.

For a factor $1 + t/a$, its tilt at r is the generating function of a Bernoulli variable of parameter $r/(a + r)$. Thus the normalized coefficient distribution of G_ϵ is a countable sum of independent Bernoulli variables. More explicitly, with independent parameters $p_j = r/(a_j + r)$ over all reciprocal-zero factors, put $S = \sum_j X_j$. Then $\sum_j p_j < \infty$, S is finite almost surely, and

$$\mathbb{E} z^S = \frac{G_\epsilon(rz)}{G_\epsilon(r)}, \quad [t^m] G_\epsilon(t) = G_\epsilon(r) r^{-m} \mathbb{P}(S = m). \quad (16.14)$$

At the saddle, $\mathbb{E} S = m$. In particular, its variance does not exceed its mean. The lower bound follows factor by factor from a further special property of the Bessel and parity factors.

Lemma 16.4 (Variance at least half the mean). *For $Q = F, H_0, H_1$ and $t > 0$, put $\mu_Q(t) = \mathcal{D} \log Q(t)$ and $b_Q(t) = \mathcal{D} \mu_Q(t)$. Then $b_Q(t) \geq \mu_Q(t)/2$.*

Proof. For F , set $z = 2\sqrt{t}$ and $R(z) = I_1(z)/I_0(z)$. Then

$$\mu_F(t) = \frac{z}{2}R(z), \quad b_F(t) = \frac{\mu_F(t)}{2} + \frac{z^2}{4}R'(z).$$

The integral representation $I_0(z) = \pi^{-1} \int_0^\pi e^{z \cos \theta} d\theta$ [19] identifies $R(z)$ with the expectation of $\cos \theta$ under the density proportional to $e^{z \cos \theta}$. Differentiating that expectation shows $R'(z) = \text{Var}_z(\cos \theta) \geq 0$.

For the parity factors, let $x = \sqrt{t}$. Direct differentiation gives

$$\begin{aligned} \mu_{H_0} &= \frac{x \tanh x}{2}, & b_{H_0} - \frac{\mu_{H_0}}{2} &= \frac{x^2 \text{sech}^2 x}{4}, \\ \mu_{H_1} &= \frac{x \coth x - 1}{2}, & b_{H_1} - \frac{\mu_{H_1}}{2} &= \frac{1 - x^2 \text{csch}^2 x}{4}. \end{aligned}$$

The last numerator is nonnegative because $\sinh x \geq x$. All formulas extend by continuity at zero. $\square$

Proof of Theorem 16.2 and Corollary 16.3. The mean of each nonconstant factor increases continuously from zero to infinity: this follows from the displayed formulas, or from its Bernoulli product and the presence of infinitely many positive Bernoulli parameters approaching one as $r \rightarrow \infty$. The total mean is strictly increasing, so (16.4) has exactly one positive solution for every $m > 0$. Variances and means add under the product. Lemma 16.4 and the Bernoulli upper bound give (16.8).

Theorem 17.1 below gives the coefficient expansion with error $O_K(B^{-K-1})$ and corrections (16.7). Since $B \geq m/2$, all its constants are independent of the walk parameters. Multiplication by the exact factor $N!\lambda^\epsilon$ proves the theorem. The explicit estimate in Proposition 17.3 gives the corollary. $\square$

16.4 Why resolving parity changes the uniformity problem

In the original variable z , the factor $I_0(2z)^D$ is even. In sparse regimes the two arcs near positive and negative real z can have comparable size, and the odd answer is a small difference of large contributions. An error controlled relative to either arc is then insufficient for an odd relative asymptotic. In (16.3), both parity factors have positive coefficients and the tilted lattice has span one. The global characteristic-function bound in the next section gives one uniform estimate for both cases.

For example, with $\lambda = 1$ and $D \gg m^2$, the first equivalents are

$$A_{2m,D}(1) \sim \frac{(2m)!D^m}{m!}, \quad A_{2m+1,D}(1) \sim \frac{(2m+1)!D^m}{m!}.$$

The typical odd bridge has one idle step. This leading behavior is automatically retained by the factorization, including when the odd idle activity becomes arbitrarily small.

16.5 A direct extension to prescribed endpoints

The same argument already covers arbitrary endpoints, including endpoints whose coordinates grow with the length. The parameter controlling the relative error is the number of available pairs after the mandatory displacement has been removed.

Let $W_{N,x}$ be the total activity of length- N walks from 0 to $x \in \mathbb{Z}^D$, with the directional and idle activities used above. Put

$$x_i^+ = \max(x_i, 0), \quad x_i^- = \max(-x_i, 0), \quad \nu_i = |x_i|, \quad h = |x|_1 = \sum_i \nu_i, \quad c_i = a_i b_i.$$

If $N < h$, then $W_{N,x} = 0$. Otherwise write $N - h = 2m + \epsilon$, where $m \geq 0$ and $\epsilon \in \{0, 1\}$, and define

$$F_\nu(t) = \nu! \sum_{k \geq 0} \frac{t^k}{k!(k + \nu)!}, \quad G_{\epsilon,x}(t) = H_\epsilon(\lambda^2 t) \prod_{i=1}^D F_{\nu_i}(c_i t).$$

In particular $F_0 = F$ and $F_\nu(0) = 1$. For $t > 0$, $F_\nu(t) = \nu! t^{-\nu/2} I_\nu(2\sqrt{t})$, but the entire series is the definition at zero.

Corollary 16.5 (Uniform prescribed-endpoint expansion). *For all nonnegative activities,*

$$W_{N,x} = P_{N,x}[t^m] G_{\epsilon,x}(t), \quad P_{N,x} = N! \lambda^\epsilon \prod_{i=1}^D \frac{a_i^{x_i^+} b_i^{x_i^-}}{\nu_i!}. \quad (16.15)$$

Powers of exponent zero in the prefactor equal one, also when the activity is zero.

Suppose $m > 0$, $P_{N,x} > 0$, and $G_{\epsilon,x}$ is nonconstant. Let $r > 0$ satisfy $\mathcal{D} \log G_{\epsilon,x}(r) = m$, and set

$$\kappa_j = \mathcal{D}^j \log G_{\epsilon,x}(r), \quad B = \kappa_2, \quad L_{N,x} = \frac{P_{N,x} G_{\epsilon,x}(r)}{r^m \sqrt{2\pi B}}.$$

Then $m/2 \leq B \leq m$, and the expansion in Theorem 16.2, with the same corrections E_s and constants depending only on its fixed order, holds for $W_{N,x}/L_{N,x}$. Moreover, for every such $m > 0$,

$$\left| \frac{W_{N,x}}{L_{N,x}} - 1 \right| \leq \frac{12}{B} \leq \frac{24}{m}.$$

Uniformity includes arbitrary variation of x, D and all activities; there is no upper restriction on any endpoint order ν_i .

Proof. For axis i , the two directional counts are $k_i + x_i^+$ and $k_i + x_i^-$ for some $k_i \geq 0$. If there are $j = 2\ell + \epsilon$ idle steps, the length constraint is $\ell + \sum_i k_i = m$. The weight of all orders of these steps is

$$\frac{N! \lambda^{2\ell+\epsilon} \prod_i a_i^{k_i+x_i^+} b_i^{k_i+x_i^-}}{(2\ell+\epsilon)! \prod_i k_i! (k_i + \nu_i)!}.$$

Summing proves (16.15).

For each nonnegative integer ν , the Bessel product [17] gives

$$F_\nu(t) = \prod_{k \geq 1} \left(1 + \frac{4t}{j_{\nu,k}^2} \right),$$

where $j_{\nu,k}$ are the positive zeros of J_ν and $\sum_k j_{\nu,k}^{-2} < \infty$. Consequently every tilted factor is again a countable Bernoulli sum. It remains only to verify the variance lower bound uniformly in ν .

Let $z = 2\sqrt{t}$ and $M_\nu(z) = F_\nu(z^2/4)$. The positive integral representation [19, Eq. 10.32.2] reads

$$M_\nu(z) = \frac{\Gamma(\nu+1)}{\sqrt{\pi} \Gamma(\nu + \frac{1}{2})} \int_{-1}^1 e^{zu} (1-u^2)^{\nu-1/2} du.$$

At $z = 0$ this normalizes a probability density. Under its exponential tilt at $z > 0$, put $R_\nu(z) = (\log M_\nu)'(z)$. Differentiating the integral gives

$$R_\nu(z) = \mathbb{E}_{\nu,z} u, \quad R'_\nu(z) = \text{Var}_{\nu,z}(u) \geq 0.$$

Therefore the mean and variance of the coefficient tilt of F_ν satisfy

$$\mu_\nu(t) = \frac{z}{2} R_\nu(z), \quad b_\nu(t) = \frac{\mu_\nu(t)}{2} + \frac{z^2}{4} R'_\nu(z) \geq \frac{\mu_\nu(t)}{2}.$$

This is the same bound as for F_0 , with no constant depending on ν . Zero activities contribute constant factors and zero mean and variance. The means and variances add, and the parity factor has already been treated. Each nonconstant factor has infinitely many Bernoulli factors, so its mean increases strictly from zero to infinity. This proves existence and uniqueness of the saddle and the variance sandwich. Theorem 17.1 and Proposition 17.3, followed by multiplication by the exact prefactor, prove the asserted expansion and envelope. $\square$

The excluded boundaries are exact, not asymptotic exceptions. If $P_{N,x} = 0$, a required directional or idle step has zero activity and $W_{N,x} = 0$. If $G_{\epsilon,x} \equiv 1$, its coefficient is zero for $m > 0$. For $m = 0$, its constant coefficient is one and $W_{N,x} = P_{N,x}$, whether or not the generator is constant. When a prefactor vanishes but the generator is nonconstant, the normalized coefficient $[t^m]G_{\epsilon,x}$ still has the same uniform expansion; no ratio to a zero walk weight is asserted.

[write] Checked, 7 October 2026. Read line by line, with no error found. Rederived by hand: the exact identity (16.3) and its endpoint form (16.15); the three formulas of Lemma 16.4 (with $\mathcal{D} = \frac{z}{2} \frac{d}{dz}$ for F and $\mathcal{D} = \frac{x}{2} \frac{d}{dx}$ for H_0, H_1); $B \leq m$ from the Bernoulli representation; and the normalization of the integral representation of F_ν (at $z = 0$ it is $F_\nu(0) = 1$). The DLMF product formulas were not read; the products are classical. The corrections (16.7) are Part I's saddle functionals: E_s is $\mathcal{T}_s 1$ of (11) (only $j = 0$ survives for a constant amplitude) with (B, κ_j) in place of (b, κ_h) ; in particular (16.9) is the last term of (24). On an anisotropic example computed exactly by the write ($D = 3$, $c = (1, \frac{1}{7}, 5)$, $\lambda = \frac{1}{3}$, $N = 10, 11, 30, 31, 60, 61$), B/m lies between 0.51 and 0.60, $|W_N/L_N - 1|$ between $1.2 \cdot 10^{-3}$ and $8.1 \cdot 10^{-3}$, far inside $12/B$ (from 4.1 down to 0.76), and, after E_1 and after $E_1 + E_2$, at most $2.0 \cdot 10^{-4}$ and $2.3 \cdot 10^{-5}$. How Theorem 16.2 relates to Part I's Theorem 9.1 is in the note at the end of Section 9.

17 A uniform local expansion for Bernoulli sums

This section proves the probabilistic estimate used in Theorem 16.2. Its hypotheses allow arbitrarily unequal Bernoulli parameters and countably many summands. The method is classical; the proof records the uniform constants and the cancellation needed for an integer-power remainder at every order.

Theorem 17.1 (Uniform Poisson–binomial local expansion). *Let X_j be a finite or countable family of independent Bernoulli random variables with parameters $p_j \in [0, 1]$ and $\sum_j p_j < \infty$. Suppose $S = \sum_j X_j$ has integer mean m and variance $B > 0$. Write κ_j for its cumulants. For every fixed $K \geq 0$, as $B \rightarrow \infty$,*

$$\sqrt{2\pi B} \mathbb{P}(S = m) = \sum_{s=0}^K E_s + O_K(B^{-K-1}), \quad (17.1)$$

where E_s is given by (16.7), with these cumulants and variance. The error constant depends only on K . In particular, $E_s = O_s(B^{-s})$ uniformly over all such families.

17.1 Analytic bounds and localization

For one Bernoulli parameter define the centered logarithm

$$\ell_p(z) = \log(1 - p + pe^{iz}) - ipz.$$

On $|z| \leq 1/4$, the logarithm is the analytic branch at zero:

$$|1 - p + pe^{iz}| \geq 1 - |e^{iz} - 1| \geq 2 - e^{1/4} > 0.$$

Direct differentiation gives

$$\ell_p''(z) = -\frac{p(1-p)e^{iz}}{(1-p+pe^{iz})^2}.$$

It is bounded there by an absolute constant times $p(1-p)$. Cauchy's estimates on a smaller fixed disk give, for each fixed $j \geq 2$,

$$|\kappa_j| \leq C_j B. \quad (17.2)$$

They also give, for each fixed $L \geq 2$ and real $|\theta| \leq \delta$,

$$\log \phi(\theta) = -\frac{B\theta^2}{2} + \sum_{j=3}^L \frac{\kappa_j(i\theta)^j}{j!} + O_L(B|\theta|^{L+1}), \quad \phi(\theta) = \mathbb{E}e^{i\theta(S-m)}. \quad (17.3)$$

Here $\delta > 0$ is a fixed sufficiently small number. These statements follow by summing the individual estimates; the summability of the variances justifies this also for countably many factors.

For the whole Fourier interval, the elementary identity

$$|1 - p + pe^{i\theta}|^2 = 1 - 4p(1-p)\sin^2(\theta/2)$$

and $1 - u \leq e^{-u}$ imply

$$|\phi(\theta)| \leq e^{-2B\sin^2(\theta/2)} \leq e^{-2B\theta^2/\pi^2}, \quad |\theta| \leq \pi. \quad (17.4)$$

Thus the characteristic function has no competing lattice peak when B grows, regardless of how close individual parameters are to zero or one. Fourier inversion at the integer mean reads

$$\mathbb{P}(S = m) = \frac{1}{2\pi} \int_{-\pi}^{\pi} \phi(\theta) d\theta. \quad (17.5)$$

17.2 Proof to every fixed order

Proof of Theorem 17.1. Fix K and take $L = 2K + 3$ in (17.3). Put $h = B^{-1/2}$, $x = \theta/h$, and $b_j = \kappa_j/B$. Then

$$\log \phi(hx) = -x^2/2 + Q(h, x) + O_K(h^{2K+2}|x|^{2K+4}), \quad Q(t, x) = \sum_{j=3}^{2K+3} \frac{b_j(ix)^j}{j!} t^{j-2}.$$

The b_j are bounded solely in terms of j . Choose $\delta > 0$ sufficiently small in terms of K . On $|hx| \leq \delta$, for $0 \leq t \leq h \leq 1$, each term in Q is bounded by a constant times $x^2(t|x|)^{j-2}$. Consequently

$$|Q(t, x)| \leq x^2/8, \quad |\log \phi(hx) + x^2/2 - Q(h, x)| \leq x^2/8.$$

Replacing $\phi(hx)$ by $e^{-x^2/2}e^{Q(h,x)}$ therefore has integrated error $O_K(h^{2K+2})$: the pointwise error is bounded by

$$C_K h^{2K+2} |x|^{2K+4} e^{-x^2/4}.$$

Expand $e^{Q(t,x)}$ in t at zero through degree $2K+1$. The integral form of Taylor's remainder, together with the preceding bound on Q , shows that its remainder after multiplication by $e^{-x^2/2}$ is at most

$$C_K h^{2K+2} (1 + |x|^{M_K}) e^{-3x^2/8}.$$

Indeed, a fixed number of derivatives of e^Q is e^Q times a polynomial in the derivatives of Q , whose coefficients are uniformly bounded for $0 \leq t \leq h \leq 1$. The displayed bound is integrable with a constant depending only on K .

The coefficient of t^w is

$$A_w(x) = \sum_{\sum (j-2)\nu_j = w} \prod_j \frac{\{b_j(ix)^j/j!\}^{\nu_j}}{\nu_j!}.$$

If $J = \sum j\nu_j$, then $J \equiv w \pmod{2}$. Hence $A_w(x)$ is odd and purely imaginary when w is odd. Its integral against the even Gaussian over the symmetric local interval is exactly zero. For even $w = 2s$, extend the polynomial Gaussian integral to the whole real line, with exponentially small error. Equation (17.4) bounds the remaining original integral by an exponentially small quantity as well.

Finally,

$$\frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} x^{2j} e^{-x^2/2} dx = (2j-1)!!.$$

Multiplying by h^{2s} and replacing b_j by κ_j/B gives exactly (16.7). For any summand in that formula,

$$J/2 - \sum_j \nu_j = \frac{1}{2} \sum_j (j-2)\nu_j = s,$$

so (17.2) gives $E_s = O_s(B^{-s})$. All discarded terms have integrated size $O_K(B^{-K-1})$ in the normalization of (17.1), completing the proof. $\square$

Remark 17.2 (Why the odd term must be retained). After the first correction, an absolute bound on an omitted fifth cumulant would yield only $O(B^{-3/2})$. The proof keeps that term until its odd integral cancels. At order K it keeps the corresponding odd degree $2K+1$ in h . This supplies the claimed $O(B^{-K-1})$ remainder without an unproved parity improvement.

17.3 A fully explicit leading bound

Proposition 17.3. *Under the hypotheses of Theorem 17.1, for every $B > 0$,*

$$\left| \sqrt{2\pi B} \mathbb{P}(S = m) - 1 \right| \leq \frac{12}{B}. \quad (17.6)$$

Proof. For $|\theta| \leq 1/2$, write $\log \phi(\theta) = -B\theta^2/2 + R(\theta) + iI(\theta)$. We first prove

$$|R(\theta)| \leq B\theta^4/8, \quad |I(\theta)| \leq B|\theta|^3/3. \quad (17.7)$$

For one parameter put $q = p(1-p)$ and $u = 4q \sin^2(\theta/2)$. Then $u \leq 1/16$, and the real remainder is

$$q\{\theta^2/2 - 2\sin^2(\theta/2)\} + \frac{1}{2}\{\log(1-u) + u\}.$$

The first term has absolute value at most $q\theta^4/24$. For the second,

$$\frac{1}{2}|\log(1-u)+u| \leq \frac{u^2}{4(1-u)} \leq \frac{4}{15}q^2\theta^4 \leq \frac{1}{15}q\theta^4.$$

Their sum is less than $q\theta^4/8$. For the imaginary part,

$$\frac{d^3}{d\theta^3} \log(1-p+pe^{i\theta}) = -\frac{iqe^{i\theta}(1-p-pe^{i\theta})}{(1-p+pe^{i\theta})^3}.$$

On this real interval its modulus is at most $2q$, since the numerator apart from q has modulus at most one and the denominator has modulus at least $\cos^3(1/4) > 1/2$. The centered first derivative vanishes and the second derivative at zero is real. Taylor's theorem gives the imaginary bound $q|\theta|^3/3$. Summing proves (17.7).

Put $a = 15/32$. The real exponent is at most $-aB\theta^2$. Using the real mean-value inequality for exponentials and $1 - \cos I \leq I^2/2$, obtain

$$|\Re\phi(\theta) - e^{-B\theta^2/2}| \leq e^{-aB\theta^2} \left(\frac{B\theta^4}{8} + \frac{B^2\theta^6}{18} \right).$$

Integrating over the entire real line and multiplying by $\sqrt{B/(2\pi)}$ bounds the local normalized error by

$$\frac{1}{B} \left\{ \frac{3}{32\sqrt{2}a^{5/2}} + \frac{5}{48\sqrt{2}a^{7/2}} \right\} < \frac{2}{B}.$$

For example, $a > 4/9$ and $1/\sqrt{2} < 3/4$ bound the braces by $45927/24576 < 2$.

On $1/2 \leq |\theta| \leq \pi$, use (17.4) and enlarge to two infinite Gaussian tails. The elementary inequality

$$\int_h^\infty e^{-cB\theta^2} d\theta \leq \frac{e^{-cBh^2}}{2cBh}$$

shows that the normalized characteristic-function tail is at most $\pi^2 e^{-B/(2\pi^2)}/\sqrt{2\pi B}$. The omitted Gaussian tail is at most $4e^{-B/8}/\sqrt{2\pi B}$. After multiplication by B , the suprema of these bounds over $B > 0$ are, respectively,

$$\frac{\pi^3}{\sqrt{2\pi e}} < 8, \quad \frac{8}{\sqrt{2\pi e}} < 2.$$

The first inequality follows from $\pi < 22/7$ and $e > 8/3$; the second from $\pi > 3$ and $e > 8/3$. Combining the local and tail estimates proves (17.6). The argument imposed no lower restriction on B . $\square$

For the lattice-bridge product, $B \geq m/2$, so this proposition gives Corollary 16.3. The numerical constant 24 is an effective certificate; its optimal value is an open quantitative question discussed in Section 21.

[write] Checked, 7 October 2026. Read line by line, with no error found. The constants of Proposition 17.3 were recomputed by hand: $|1-p+pe^{i\theta}|^2 = 1-4p(1-p)\sin^2(\theta/2)$; $u \leq q\theta^2 \leq 1/16$; $\frac{1}{24} + \frac{1}{15} = \frac{13}{120} < \frac{1}{8}$; the third derivative displayed, with $|1-p+pe^{i\theta}| \geq \cos(1/4)$ and $\cos^3(1/4) = 0.909\dots > \frac{1}{2}$; $a = \frac{1}{2} - \frac{1}{32} = \frac{15}{32}$; the Gaussian moments $\frac{3}{4}\sqrt{\pi}(aB)^{-5/2}$ and $\frac{15}{8}\sqrt{\pi}(aB)^{-7/2}$, which give the displayed braces, $1.485\dots$ at $a = 15/32$ and at most $45927/24576 = 1.868\dots$ with $a > 4/9$; the two tails, whose suprema over B are attained at $B = \pi^2$ and $B = 4$ and equal $\pi^{5/2}/\sqrt{2e} = 7.502\dots$ and $8/\sqrt{2\pi e} = 1.935\dots$; and $2 + 8 + 2 = 12$.

18 The sparse collision and idle-step transition

Let $N = 2m + \epsilon$, where $\epsilon \in \{0, 1\}$, and let $D \geq 1$. A bridge is an N -step closed walk in $\mathbb{Z}^D$ with steps $0, \pm e_1, \dots, \pm e_D$. Write J for its number of idle steps. For each axis i , closure forces the positive and negative step counts to agree; call their common value K_i . Thus

$$J \equiv \epsilon \pmod{2}, \quad \frac{J - \epsilon}{2} + \sum_{i=1}^D K_i = m.$$

Let

$$R = \#\{i : K_i \geq 2\}, \quad U = \#\{i : K_i \geq 3\}, \quad K_{\text{occ}} = \#\{i : K_i \geq 1\}.$$

Here a repeated axis means an axis used at least twice in each direction. This is a count of axes, not a count of individual step coincidences.

Give a bridge weight λ^J , where $\lambda \geq 0$, and denote the total weight by $A_{N,D}(\lambda)$. Define the entire functions

$$F(t) = \sum_{k \geq 0} \frac{t^k}{(k!)^2} = I_0(2\sqrt{t}), \quad H_0(t) = \sum_{h \geq 0} \frac{t^h}{(2h)!}, \quad H_1(t) = \sum_{h \geq 0} \frac{t^h}{(2h+1)!}.$$

The square-root notation is only shorthand for these entire series. The exact parity-reduced enumeration is

$$A_{N,D}(\lambda) = N! \lambda^\epsilon [t^m] F(t)^D H_\epsilon(\lambda^2 t). \quad (18.1)$$

Indeed, a fixed vector of axis counts and idle count contributes the multinomial number $N!/(J! \prod_i (K_i!)^2)$.

When $\epsilon = 1$ and $\lambda = 0$, the total weight in (18.1) is zero, so it cannot normalize a probability law. We therefore define the normalized partition function by the entire extension

$$\tilde{A}_{N,D}(\lambda) := N! [t^m] F(t)^D H_\epsilon(\lambda^2 t).$$

For $\lambda > 0$ this equals $A_{N,D}(\lambda)/\lambda^\epsilon$. At $\lambda = 0$, its associated limiting probability law is the uniform law on bridges with exactly ϵ idle steps. In particular it is the usual nonlazy bridge law for $\epsilon = 0$, and the exactly-one-idle bridge law for $\epsilon = 1$. Assertions about the odd law at zero below refer to this extension, not to normalization of zero total weight.

The two transition parameters are

$$\theta = \frac{m^2}{D}, \quad \beta = \lambda \sqrt{\frac{m}{D}}.$$

The first controls repeated axes; the second controls the idle count. The original unweighted bridge measure $\lambda = 1$ has $\beta \rightarrow 0$ throughout the regime of bounded θ considered here. A nonzero idle transition is observed by allowing the idle weight to grow.

Theorem 18.1 (Uniform sparse partition function). *Fix $K, B < \infty$. Uniformly over integers $D \geq 1$, parities $\epsilon \in \{0, 1\}$, and $\lambda \geq 0$ such that $0 < \theta \leq K$ and $0 \leq \beta \leq B$, as $m \rightarrow \infty$,*

$$\tilde{A}_{N,D}(\lambda) = \frac{N! D^m}{m!} e^{-\theta/4} H_\epsilon(\beta^2) \left(1 + \frac{C_\epsilon(\theta, \beta)}{m} + O_{K,B}(m^{-2}) \right), \quad (18.2)$$

where, with $c = \beta^2$,

$$C_\epsilon(\theta, \beta) = \frac{\theta}{4} - \frac{\theta^2}{72} + \frac{\theta}{2} \frac{c H'_\epsilon(c)}{H_\epsilon(c)} - \frac{1}{2} \frac{c^2 H''_\epsilon(c)}{H_\epsilon(c)}. \quad (18.3)$$

Multiplication by λ^ϵ gives the weighted count, with the zero odd count correctly retained when $\lambda = 0$.

Define a probability law $\mathcal{P}_{\epsilon,\beta}$ on the nonnegative integers by its probability generating function

$$P_{\epsilon,\beta}(u) = \frac{u^\epsilon H_\epsilon(\beta^2 u^2)}{H_\epsilon(\beta^2)}. \quad (18.4)$$

For $\beta > 0$ this is $\text{Poisson}(\beta)$ conditioned to have parity ϵ : it is $\cosh(\beta u)/\cosh \beta$ in the even case, and $\sinh(\beta u)/\sinh \beta$ in the odd case. At $\beta = 0$ it is respectively the point mass at 0 or at 1. This definition avoids conditioning a zero-parameter Poisson variable on an event of probability zero.

Theorem 18.2 (Joint sparse probability law). *Under the same parameter restrictions, let $p_{m,D,\lambda}(j,r)$ be the joint law of (J,R) under the weighted bridge measure, including the minimal-idle extension at zero. For every fixed $a, b \geq 1$,*

$$\sum_{j,r \geq 0} a^j b^r \left| p_{m,D,\lambda}(j,r) - \mathcal{P}_{\epsilon,\beta}\{j\} e^{-\theta/4} \frac{(\theta/4)^r}{r!} \right| = O_{K,B,a,b}(m^{-1}). \quad (18.5)$$

Consequently the idle count and the repeated-axis count are asymptotically independent, uniformly in these moving parameters. If $\theta \rightarrow \theta_*$ and $\beta \rightarrow \beta_*$ along a fixed parity, their joint limit is $\mathcal{P}_{\epsilon,\beta_*} \otimes \text{Pois}(\theta_*/4)$. Moreover

$$\mathbb{E}U = \frac{\theta^2}{36m} + O_{K,B}(m^{-2}), \quad \mathbb{P}(U \geq 1) = O_{K,B}(m^{-1}). \quad (18.6)$$

If $Q \sim \mathcal{P}_{\epsilon,\beta}$ and $Z \sim \text{Pois}(\theta/4)$ are independent, then

$$d_{\text{TV}}(\mathcal{L}(m - K_{\text{occ}}), \mathcal{L}((Q - \epsilon)/2 + Z)) = O_{K,B}(m^{-1}). \quad (18.7)$$

Thus sparse occupied-axis fluctuations are described by an explicit discrete law, on the natural scale of one axis.

18.1 A uniform coefficient-extraction lemma

For a function f analytic on a neighborhood of the closed disk $|z| \leq R$, with fixed $R > 1$, define

$$T_m f := \frac{m!}{m^m} [z^m] e^{mz} f(z).$$

Writing $f(z) = \sum_{j \geq 0} f_j z^j$ gives the finite exact identity

$$T_m f = \sum_{j=0}^m f_j \frac{(m)_j}{m^j}. \quad (18.8)$$

Here $(m)_j = m(m-1) \cdots (m-j+1)$ is the falling factorial. For $0 \leq j \leq m$, the elementary product inequalities give

$$0 \leq 1 - \frac{(m)_j}{m^j} \leq \frac{j(j-1)}{2m}.$$

Cauchy's estimate $|f_j| \leq \|f\|_R R^{-j}$ and the exponentially small tail $j > m$ show

$$T_m f = f(1) + O_R(m^{-1} \|f\|_R). \quad (18.9)$$

For a second order bound, apply the first two Bonferroni inequalities to the product $\prod_{k=0}^{j-1} (1 - k/m)$:

$$0 \leq \frac{(m)_j}{m^j} - 1 + \frac{j(j-1)}{2m} \leq \frac{1}{2} \left(\frac{j(j-1)}{2m} \right)^2.$$

Summing again yields the uniform identity

$$T_m f = f(1) - \frac{f''(1)}{2m} + O_R(m^{-2}\|f\|_R). \quad (18.10)$$

The missing tails of both $f(1)$ and $f''(1)$ are exponentially small. These estimates also apply to analytic families with a common sup norm bound. This elementary lemma is what supplies uniformity for the changing dimension; no uncontrolled saddle substitution is used.

18.2 Proof of the count and joint law

To mark repeated axes, put

$$F_w(t) = 1 + t + w(F(t) - 1 - t).$$

The exact joint generating polynomial, after removal of the common λ^ϵ factor, is

$$N!u^\epsilon[t^m]F_w(t)^D H_\epsilon(\lambda^2 u^2 t). \quad (18.11)$$

This formula is polynomial in u, w, λ , and remains meaningful at $\lambda = 0$ with the interpretation specified above.

Fix any complex disks $|u| \leq R_u$ and $|w| \leq R_w$, and a disk $|z| \leq R$ with $R > 1$. Put $t_0 = m/D = \theta/m$ and substitute $t = t_0 z$. The function $F_w(t_0 z)$ is uniformly close to 1 for large m ; its logarithm is the branch obtained by Taylor expansion at 0. Uniformly on these three fixed disks,

$$\log F_w(s) = s + \frac{w-2}{4}s^2 + \frac{3-2w}{9}s^3 + O_{R_w}(s^4).$$

Hence

$$D \log F_w(t_0 z) = mz + \frac{w-2}{4}\theta z^2 + \frac{3-2w}{9m}\theta^2 z^3 + O_{K,R,R_w}(m^{-2}). \quad (18.12)$$

The last estimate follows from $Dt_0^4 = m^4/D^3 = \theta^3/m^2$. Similarly $\lambda^2 t_0 = \beta^2$ exactly, so the idle factor is $H_\epsilon(\beta^2 u^2 z)$, which is uniformly bounded on the fixed disks because $\beta \leq B$.

Set

$$\alpha = \frac{w-2}{4}\theta, \quad b_0 = \frac{3-2w}{9}\theta^2, \quad c = \beta^2 u^2, \quad f(z) = e^{\alpha z^2} H_\epsilon(cz).$$

Equations (18.12) and (18.10) give

$$\frac{m!}{D^m}[t^m]F_w(t)^D H_\epsilon(\lambda^2 u^2 t) = f(1) + \frac{b_0 f(1) - f''(1)/2}{m} + O(m^{-2}). \quad (18.13)$$

Expanding the derivative, the right-hand side is

$$e^\alpha \left[H_\epsilon(c) + \frac{1}{m} \{ (b_0 - \alpha - 2\alpha^2) H_\epsilon(c) - 2\alpha c H'_\epsilon(c) - \frac{1}{2} c^2 H''_\epsilon(c) \} \right] + O(m^{-2}).$$

All error constants here are uniform for the stipulated real parameters and fixed complex marker disks. To justify applying the transfer to the analytic remainder, observe from (18.8) and Cauchy's bound that $|T_m g| \leq (1 - 1/R)^{-1} \|g\|_R$. Thus a remainder $O(m^{-2})$ in sup norm remains $O(m^{-2})$ after transfer.

At $u = w = 1$ one has $\alpha = -\theta/4$ and $b_0 = \theta^2/9$. The expression $b_0 - \alpha - 2\alpha^2$ then reduces to $\theta/4 - \theta^2/72$, and (18.13) proves Theorem 18.1. The leading factor cannot approach zero: $e^{-\theta/4} \geq e^{-K/4}$ and $H_\epsilon(\beta^2) \geq 1$. Division by the unmarked count is therefore uniform.

Let $G_{m,D,\lambda}(u, w)$ be the joint probability generating function of (J, R) . Taking the ratio of the marked and unmarked formulas, and restoring u^ϵ , proves

$$G_{m,D,\lambda}(u, w) = \frac{u^\epsilon H_\epsilon(\beta^2 u^2)}{H_\epsilon(\beta^2)} \exp\left(\frac{\theta}{4}(w-1)\right) + O(m^{-1}) \quad (18.14)$$

uniformly on every fixed complex polydisk in u, w . Given $a, b \geq 1$, choose $R_u > a$ and $R_w > b$. Cauchy's estimates on the larger disks give a bound $Cm^{-1}R_u^{-j}R_w^{-r}$ for each coefficient of the error. Summing $a^j b^r$ times this bound yields (18.5). This also shows why convergence of the generating functions merely on a neighborhood of 1 would be insufficient for the asserted weighted norm: the present proof supplies every fixed complex disk directly.

For the assertion about triple-use axes, introduce an extra marker v and set

$$F_{w,v}(t) = 1 + t + \frac{wt^2}{4} + wv \left(F(t) - 1 - t - \frac{t^2}{4} \right).$$

Its logarithmic coefficient of t^3 is $wv/36 - w/4 + 1/3$. Therefore the argument above, uniformly also on a fixed disk about $v = 1$, gives a v derivative at $w = v = 1$ equal to

$$\frac{\theta^2}{36m} e^{-\theta/4} H_\epsilon(\beta^2) + O_{K,B}(m^{-2})$$

for the normalized numerator. Division by the unmarked expression proves the expectation in (18.6); Markov's inequality proves its probability bound. Differentiation of the error is valid by Cauchy's estimate on a slightly larger fixed v disk.

On the event $U = 0$, every occupied axis has one or two pairs, so

$$m - K_{\text{occ}} = (J - \epsilon)/2 + R.$$

The failure probability is $O(m^{-1})$. Apply the data-processing inequality for total variation to (18.5) at $a = b = 1$ and add this failure probability. This proves (18.7) and completes the theorem.

18.3 Consequences and scope

For unweighted bridges $\lambda = 1$, $\beta^2 = m/D = \theta/m$. Thus $J = \epsilon$ with probability $1 - O_K(m^{-1})$, and the occupied-axis deficit has the Poisson($\theta/4$) approximation in total variation with error $O_K(m^{-1})$. For example, the even count is

$$A_{2m,D}(1) = \frac{(2m)! D^m}{m!} e^{-\theta/4} \left(1 + \frac{3\theta/4 - \theta^2/72}{m} + O_K(m^{-2}) \right),$$

because $H_0(\theta/m) = 1 + \theta/(2m) + O(m^{-2})$. For odd lengths,

$$A_{2m+1,D}(1) = \frac{(2m+1)! D^m}{m!} e^{-\theta/4} \left(1 + \frac{5\theta/12 - \theta^2/72}{m} + O_K(m^{-2}) \right),$$

using $H_1(\theta/m) = 1 + \theta/(6m) + O(m^{-2})$. These statements remain uniform as $\theta \downarrow 0$.

The theorem addresses the critical sparse range $m^2/D = O(1)$. It does not itself describe a diverging repeated-axis mean $m^2/D \rightarrow \infty$. The global Bernoulli-product saddle theorem can cover counts in that wider range, but a Gaussian or mod-Poisson description of the repeated-axis law would require additional uniform marked estimates.

[write] Checked, 7 October 2026. Read line by line, with no error found. Rederived by hand: the transfer inequalities; the coefficients $(w-2)/4$ and $(3-2w)/9$ of $\log F_w$ and $Dt_0^k = m, \theta, \theta^2/m, \theta^3/m^2$ for $k = 1, \dots, 4$; the second derivative of f at 1 and $b_0 - \alpha - 2\alpha^2 = \theta/4 - \theta^2/72$ at $u = w = 1$, hence (18.3); the triple-use coefficient $wv/36 - w/4 + 1/3$; the identity $m - K_{\text{occ}} = (J - \epsilon)/2 + R$ on $\{U = 0\}$; and the unweighted corrections $3\theta/4 - \theta^2/72$ and $5\theta/12 - \theta^2/72$. From exact rational counts of its own (a binary power of the truncated series of F), the write reproduced all sixteen entries of Table 4 to their printed digits. Further cases: at $\theta = 4, \beta = 2$ ($D = m^2/4$) and $m = 20, 40, 80$ the corrected relative errors are $-2.9 \cdot 10^{-3}, -8.6 \cdot 10^{-4}, -2.3 \cdot 10^{-4}$ for even N and $-2.4 \cdot 10^{-3}, -6.7 \cdot 10^{-4}, -1.8 \cdot 10^{-4}$ for odd N , of order m^{-2} ; at $\theta = \frac{1}{4}, \beta = 0$ ($D = 4m^2$) they are $-1.9 \cdot 10^{-6}, -4.5 \cdot 10^{-7}, -1.1 \cdot 10^{-7}$ for both parities. For the unweighted measure, $m \mathbb{E}U$ is 0.0243 (even N) and 0.0244 (odd) at $\theta = 1, m = 20, 0.0261$ for both at $m = 40$, and 0.433, 0.434 at $\theta = 4, m = 40$, against $\theta^2/36 = 0.0277\dots$ and $0.444\dots$, with the gap shrinking like $1/m$.

19 The Gaussian transition and its dimension thresholds

For integers $N, D \geq 1$, let

$$R_{N,D} = \frac{N!}{(2D+1)^N} [z^N] e^z I_0(2z)^D, \quad G_{N,D} = \left(\frac{2D+1}{4\pi N} \right)^{D/2}.$$

Thus $R_{N,D}$ is the return probability of the walk whose $2D+1$ equiprobable increments are $0, \pm e_1, \dots, \pm e_D$. The covariance matrix of one increment is $2(2D+1)^{-1}I_D$, so $G_{N,D}$ is its Gaussian local-limit prediction. Put

$$\rho = \frac{N}{D}, \quad q = \rho^{-1} = \frac{D}{N}.$$

The result is uniform in the two integer parameters; in particular, it does not first fix the dimension.

Theorem 19.1 (Dense expansion and Gaussian threshold). *There are absolute constants C and ρ_0 such that, for $D \geq 1$ and $N/D \geq \rho_0$,*

$$\log \frac{R_{N,D}}{G_{N,D}} = -\frac{3D}{16N} + \frac{D^3}{48N^2} + O\left(\frac{D^2}{N^2} + \frac{D^4}{N^3} + \frac{1}{N} + e^{-N/(4D)}\right). \quad (19.1)$$

The implied constant is absolute. Consequently, along every sequence with $N/D \rightarrow \infty$:

1. $R_{N,D}/G_{N,D} \rightarrow 1$ if and only if $D^3/N^2 \rightarrow 0$;
2. if $D^3/N^2 \rightarrow \xi \in (0, \infty)$, then $R_{N,D}/G_{N,D} \rightarrow e^{\xi/48}$;
3. if $D^3/N^2 \rightarrow \infty$, then $\log(R_{N,D}/G_{N,D}) \sim D^3/(48N^2)$.

In particular, the critical scale is $N \asymp D^{3/2}$. If $N/D^{3/2} \rightarrow s \in (0, \infty)$, the critical ratio is $\exp(1/(48s^2))$. The same conclusions hold for even and odd N .

Proof. We first convert the parity-resolved saddle into a saddle involving $e^z I_0(2z)^D$. This conversion is justified below; it is not an assumption that a second saddle can be discarded uniformly.

Write $N = 2m + \epsilon$, with $\epsilon \in \{0, 1\}$, and set

$$F(t) = I_0(2\sqrt{t}), \quad H_0(t) = \cosh \sqrt{t}, \quad H_1(t) = \frac{\sinh \sqrt{t}}{\sqrt{t}}.$$

The exact parity identity is

$$[z^N]e^z I_0(2z)^D = [t^m]F(t)^D H_\epsilon(t).$$

Theorem 16.2 for this coefficient has relative error $O(1/m) = O(1/N)$, with an absolute constant.

Define

$$B(z) = z \frac{d}{dz} \log I_0(2z), \quad A_+(z) = DB(z) + z,$$

and let $\zeta > 0$ solve $A_+(\zeta) = N$. Existence and uniqueness follow from positivity of the tilted variance: $A'_+(z) > 0$; also $A_+(0) = 0$ and $A_+(z) \rightarrow \infty$. Set

$$V_+ = \zeta A'_+(\zeta).$$

We claim the following uniformly valid positive-saddle formula:

$$R_{N,D} = \frac{N! e^\zeta I_0(2\zeta)^D}{(2D+1)^N \zeta^N \sqrt{2\pi V_+}} \left(1 + O\left(N^{-1} + e^{-N/(4D)}\right) \right). \quad (19.2)$$

For completeness, put

$$Q_\epsilon(z) = 1 + (-1)^\epsilon e^{-2z}.$$

Then $H_\epsilon(z^2) = e^z Q_\epsilon(z)/(2z^\epsilon)$. If η^2 is the exact parity saddle, its equation is

$$A_+(\eta) + \eta \frac{Q'_\epsilon(\eta)}{Q_\epsilon(\eta)} = N. \quad (19.3)$$

The standard large-positive-argument expansion of I_0 and its derivatives implies

$$B(z) = 2z - \frac{1}{2} + O(z^{-1}), \quad B'(z) = 2 + O(z^{-2}), \quad B''(z) = O(z^{-3}).$$

Hence $\zeta \asymp \rho$, $A'_+(z) \asymp D$ near ζ , and, in fact, $\zeta \geq N/(2D+1) \geq \rho/3$ exactly, since $I_1(2\zeta)/I_0(2\zeta) < 1$ by the positive integral representation. For $z \geq 1$ the perturbing term in (19.3) is $O(ze^{-2z})$, and its derivative is $O(ze^{-2z})$. The mean-value theorem, first on $[\zeta - 1, \zeta + 1]$ and then between the two roots, gives

$$|\eta - \zeta| \leq C \frac{\zeta e^{-2\zeta}}{D}.$$

The positive exponent $E(z) = D \log I_0(2z) + z - N \log z$ has $E'(\zeta) = 0$ and $E''(z) = O(D/\zeta)$ on this short interval. Its change is therefore $O(\zeta e^{-4\zeta}/D)$, and $\log Q_\epsilon(\eta) = O(e^{-2\zeta})$. If V_t denotes the variance at the exact parity saddle, direct differentiation gives

$$4V_t = \eta A'_+(\eta) + \eta \frac{d}{d\eta} \left(\eta \frac{Q'_\epsilon(\eta)}{Q_\epsilon(\eta)} \right).$$

It follows that $4V_t/V_+ = 1 + O(\zeta e^{-2\zeta})$. The factor $1/2$ in H_ϵ consequently cancels the factor 2 from replacing $\sqrt{V_t}$ by $\sqrt{V_+}/2$. Since $\zeta e^{-2\zeta} = O(e^{-\rho/4})$, the exact parity saddle theorem proves (19.2). This also explains the uniform treatment of the two parities.

It remains to expand this real saddle. Let

$$c = 1 + \frac{1}{2D} \in [1, 3/2], \quad w = 2\zeta, \quad v = cw,$$

and define

$$\ell(w) = \log I_0(w) - w + \frac{1}{2} \log(2\pi w), \quad \ell_c(v) = \ell(v/c).$$

The classical expansion [20], for positive $w \rightarrow \infty$, is

$$\ell(w) = \frac{1}{8w} + \frac{1}{16w^2} + O(w^{-3}). \quad (19.4)$$

The expansion may be differentiated to the fixed orders used here; this follows either from the differentiated Bessel expansions or from Laplace's method applied to $I_0(w) = \pi^{-1} \int_0^\pi e^{w \cos \theta} d\theta$. All resulting bounds for ℓ_c are uniform for $1 \leq c \leq 3/2$. The saddle equation becomes

$$\rho = v - \frac{1}{2} + v\ell'_c(v) = v - \frac{1}{2} - \frac{c}{8v} - \frac{c^2}{8v^2} + O(v^{-3}). \quad (19.5)$$

Its derivative with respect to v is $1 + O(v^{-2})$, uniformly in c . Substitution followed by the mean-value theorem therefore gives the uniform inverse expansion

$$v = \rho + \frac{1}{2} + \frac{c}{8\rho} + \frac{2c^2 - c}{16\rho^2} + O(\rho^{-3}). \quad (19.6)$$

Applying Stirling's formula with remainder $O(1/N)$ to (19.2), and subtracting $\log G_{N,D}$, yields

$$\log \frac{R_{N,D}}{G_{N,D}} = D\mathcal{A}(\rho, c) + \frac{1}{2} \log \frac{N}{V_+} + O(N^{-1} + e^{-\rho/4}), \quad (19.7)$$

where

$$\mathcal{A}(\rho, c) = (\rho + \frac{1}{2}) \log \frac{\rho}{v} - \rho + v + \ell_c(v).$$

There is no untracked D -dependent constant here: the factors $(2D+1)^N$, the Gaussian normalization, and the Bessel prefactor have all been retained exactly.

Using (19.6) in this last display gives

$$\mathcal{A}(\rho, c) = \frac{c-1}{8\rho} + \frac{1}{\rho^2} \left(\frac{1}{48} + \frac{c^2 - c}{16} \right) + O(\rho^{-3}). \quad (19.8)$$

Furthermore, since $V_+ = Dv \, d\rho/dv$,

$$\frac{1}{2} \log \frac{N}{V_+} = -\frac{1}{4\rho} + \frac{1}{\rho^2} \left(\frac{1}{16} - \frac{c}{8} \right) + O(\rho^{-3}).$$

Because $D(c-1) = 1/2$, the terms of order ρ^{-1} combine to $-3/(16\rho)$. The leading term of order $D\rho^{-2}$ is $D/(48\rho^2)$, and all other second-order terms are $O(\rho^{-2})$. Multiplying the uniform remainder in (19.8) by D proves (19.1).

Finally put $\chi = D/\rho^2 = D^3/N^2$. Formula (19.1) reads

$$\log(R_{N,D}/G_{N,D}) = \frac{\chi}{48} - \frac{3}{16\rho} + O\left(\frac{\chi}{\rho} + \rho^{-2} + N^{-1} + e^{-\rho/4}\right).$$

This gives the three asserted limits. For the “only if” direction in the first assertion, any failure of $\chi \rightarrow 0$ has a subsequence on which χ converges to a positive finite value or tends to infinity, or a further subsequence of one of these two forms. The other two assertions then preclude a ratio tending to one. $\square$

Corollary 19.2 (Nonlazy walks and failure of every constant Gaussian bound). *For positive even N , let*

$$R_{N,D}^0 = \frac{N!}{(2D)^N} [z^N] I_0(2z)^D, \quad G_{N,D}^0 = \left(\frac{D}{2\pi N} \right)^{D/2}.$$

Uniformly as $N/D \rightarrow \infty$,

$$\log \frac{R_{N,D}^0}{2G_{N,D}^0} = -\frac{D}{4N} + \frac{D^3}{48N^2} + O\left(\frac{D^2}{N^2} + \frac{D^4}{N^3} + \frac{1}{N}\right). \quad (19.9)$$

The three conclusions of Theorem 19.1 hold with R/G replaced by $R^0/(2G^0)$. In particular, there is no universal finite constant K for which

$$R_{N,D}^0 \leq K \left(\frac{D}{2\pi N} \right)^{D/2} \quad \text{for every positive even } N \text{ and every } D.$$

Proof. The exact parity identity now has no idle factor: $[z^{2m}] I_0(2z)^D = [t^m] F(t)^D$. Its variance in the t variable is exactly one fourth of the variance in z , yielding the lattice factor 2 in the denominator of (19.9). The saddle computation is otherwise exactly the preceding proof with $c = 1$. Thus the ρ^{-1} term in $D\mathcal{A}$ vanishes, while the prefactor contributes $-1/(4\rho)$ and $D\mathcal{A}$ contributes $D/(48\rho^2)$. There is no parity approximation and hence no exponentially small error to retain. Choosing any even N with $N \sim D^{4/3}$ gives

$$\log \frac{R_{N,D}^0}{2G_{N,D}^0} \sim \frac{D^{1/3}}{48} \rightarrow \infty,$$

which excludes every finite K . □

Corollary 19.3 (The side of approach to the Gaussian value). *If $D \rightarrow \infty$ and $N/D^2 \rightarrow s \in (0, \infty)$, then*

$$D \log \frac{R_{N,D}}{G_{N,D}} \rightarrow \frac{1 - 9s}{48s^2}.$$

For even N in the same regime,

$$D \log \frac{R_{N,D}^0}{2G_{N,D}^0} \rightarrow \frac{1 - 12s}{48s^2}.$$

Thus the eventual signs switch at $s = 1/9$ in the lazy model and $s = 1/12$ in the nonlazy model, away from the critical equality cases.

Proof. Multiply (19.1) or (19.9) by D . Every remainder is $O(D^{-1})$ or exponentially smaller in this regime. □

Corollary 19.4 (The eventual Gaussian upper-bound thresholds). *Let $T_0(D)$ be the least positive even integer M such that $R_{N,D}^0 \leq 2G_{N,D}^0$ for every positive even $N \geq M$. Let $T_1(D)$ be the least positive integer M such that $R_{N,D} \leq G_{N,D}$ for every integer $N \geq M$. These thresholds are finite, and*

$$\frac{T_0(D)}{D^2} \rightarrow \frac{1}{12}, \quad \frac{T_1(D)}{D^2} \rightarrow \frac{1}{9} \quad (D \rightarrow \infty).$$

More precisely, for each $\delta > 0$ and all sufficiently large D , the nonlazy upper bound holds simultaneously at every even $N \geq (1/12 + \delta)D^2$, and the lazy upper bound holds at every $N \geq (1/9 + \delta)D^2$. Each model has a failing time arbitrarily close to its corresponding lower scale $(1/12 - \delta)D^2$ or $(1/9 - \delta)D^2$ when that scale is positive.

Proof. For the nonlazy model multiply (19.9) by ρ . For the lazy model do the same in (19.1). Writing $a = 1/4$ or $a = 3/16$, respectively, gives

$$\rho \log(\text{return/Gaussian}) = -a + \frac{D}{48\rho} + O\left(\rho^{-1} + D\rho^{-2} + D^{-1} + \rho e^{-\rho/4}\right),$$

where the last term is omitted in the nonlazy case and its Gaussian includes the period factor. For every fixed $c > 0$, the error is $O_c(D^{-1})$ uniformly on the entire half-line $\rho \geq cD$. Thus if $c > 1/(48a)$, the expression is at most $-a + 1/(48c) + O_c(D^{-1}) < 0$ for all sufficiently large D , simultaneously for every allowed time on that half-line.

If $0 < c < 1/(48a)$, choose an allowed integer N_D nearest to cD^2 . Corollary 19.3 makes its ratio strictly greater than one for large D , so the eventual threshold exceeds N_D . Together these statements squeeze the thresholds to the asserted limits; parity rounding changes times by only $O(1)$. For each fixed dimension, finiteness in the nonlazy case follows from the eventual bound in [12]; in the lazy case it follows from the negative first fixed-dimension correction established below. $\square$

19.1 The first two fixed-dimension coefficients

The coefficient of D^3/N^2 has an independent check that also explains the cancellation of a prospective D^2/N obstruction. For fixed D , Fourier inversion gives

$$R_{N,D} = (2\pi)^{-D} \int_{[-\pi,\pi]^D} \left(\frac{1 + 2 \sum_{j=1}^D \cos \theta_j}{2D+1} \right)^N d\theta.$$

With $a = 2D+1$ and $\theta_j = \sqrt{a/(2N)} x_j$, write $S_r = \sum_{j=1}^D x_j^r$. Locally near $\theta = 0$, the exponent through order N^{-2} is

$$-\frac{S_2}{2} + \frac{A(x)}{N} + \frac{B(x)}{N^2} + O_D(N^{-3}(1 + \|x\|^8)),$$

where

$$A(x) = \frac{aS_4}{48} - \frac{S_2^2}{8}, \quad B(x) = -\frac{a^2 S_6}{2880} + \frac{aS_2 S_4}{96} - \frac{S_2^3}{24}.$$

For fixed D , the absolute characteristic function is bounded by some $q_D < 1$ outside any sufficiently small fixed neighborhood of zero: the idle step makes this walk aperiodic. On the local part, a rescaled cutoff $\|x\| \leq N^\delta$, with fixed sufficiently small $\delta > 0$, gives an integrable Gaussian Taylor remainder; the remaining local tail is exponentially small in $N^{2\delta}$. Thus integration against D independent standard Gaussian variables gives

$$\mathbb{E}A = -\frac{3D}{16}, \quad \mathbb{E}B + \frac{1}{2}\mathbb{E}A^2 = \frac{D(32D^2 - 21D - 8)}{1536}.$$

Thus, for fixed D ,

$$\begin{aligned} \frac{R_{N,D}}{G_{N,D}} &= 1 - \frac{3D}{16N} + \frac{D(32D^2 - 21D - 8)}{1536N^2} + O_D(N^{-3}), \\ \log \frac{R_{N,D}}{G_{N,D}} &= -\frac{3D}{16N} + \frac{D(4D^2 - 6D - 1)}{192N^2} + O_D(N^{-3}). \end{aligned}$$

This calculation is a check of the coefficients, not the source of the uniform growing-dimension error bound. The latter was proved by the scalar parity saddle above. In particular, directly promoting the fixed- D error term $O_D(N^{-3})$ to a uniform statement would not establish the dimension threshold.

Reference for the Bessel expansion. The only classical analytic input beyond the uniform coefficient saddle theorem is the fixed-order large-positive-argument expansion of I_0 and its derivatives, as in NIST DLMF, Section 10.40, especially equations 10.40.1 and 10.40.3, together with the integral representation in Section 10.32. Source URLs: <https://dlmf.nist.gov/10.40>, <https://dlmf.nist.gov/10.32>.

[write] **Checked, 7 October 2026.** Read line by line, with no error found. By hand: the parity identity and $H_\epsilon(z^2) = e^z Q_\epsilon(z)/(2z^\epsilon)$; $\zeta \geq N/(2D+1)$; the cancellation of the factor $\frac{1}{2}$ of H_ϵ against $\sqrt{V_t} \approx \sqrt{V_+}/2$; the two corollaries on signs and thresholds $((1-9s)/(48s^2), (1-12s)/(48s^2); 1/(48a) = 1/9 \text{ and } 1/12)$. With SymPy, keeping an unknown w^{-3} coefficient in ℓ : the inverse expansion (19.6); the coefficients $(c-1)/8$ and $\frac{1}{48} + \frac{c^2-c}{16}$ of (19.8); $\frac{1}{2} \log(N/V_+) = -\frac{1}{4\rho} + (\frac{1}{16} - \frac{c}{8})\rho^{-2} + O(\rho^{-3})$; the totals $-3/(16\rho)$ and $(\frac{D}{48} - \frac{1}{32} - \frac{3}{64D})\rho^{-2}$ for $c = 1 + \frac{1}{2D}$, and $-1/(4\rho)$, $(\frac{D}{48} - \frac{1}{16})\rho^{-2}$ for $c = 1$; the coefficients $\frac{1}{8}, \frac{1}{16}$ of ℓ from the large-argument series $1 + \frac{1}{8w} + \frac{9}{128w^2} + \dots$; and the logarithm $D(4D^2 - 6D - 1)/192$ of the fixed- D expansion. From exact counts at $D = 100$ the write obtained $R_{N,D}/G_{N,D} = 1.077912419, 1.068950263, 1.004262565, 1.004177988, 0.9960407379, 0.9960398689$ at $N = 500, 501, 1000, 1001, 2000, 2001$ (each rounded in its last digit): the $D = 100$ rows of Table 3, saddle values through E_2 , agree with these exact ratios to every printed digit. Its own saddle code reproduces all 24 entries of that table and the limits $e^{1/(48s^2)}$. For fixed D , Richardson extrapolation of $N^2(R_{N,D}/G_{N,D} - 1 + 3D/(16N))$ from exact counts at $N = 400$ and 800 gives $0.0019530\dots, 0.1015622\dots, 0.4238214\dots$ for $D = 1, 2, 3$, against $D(32D^2 - 21D - 8)/1536 = 0.0019531\dots, 0.1015625, 0.4238281\dots$: numerical corroboration of the second coefficient. The first, $-3D/16$, is the term of Part I's dated note in Remark 9.2, obtained here by the same Edgeworth computation; the lazy half of Corollary 19.4 rests on it for the finiteness of $T_1(D)$ at fixed D , with the remainder of this subsection given in outline.

20 Exact computation and reproducible checks

The proofs above establish the infinite-parameter assertions. The accompanying programs provide independent finite checks, an exact counterexample certificate, and numerical illustrations. Every large dimension entry below is identified by its evaluation method.

20.1 An integer coefficient recurrence

For isotropic walks define

$$Q_{n,D} = (n!)^2 [t^n] F(t)^D.$$

These numbers are integers, with $Q_{0,D} = 1$. The following recurrence computes all coefficients through degree m using $O(m^2)$ integer arithmetic operations, without a loop over the dimension:

$$nQ_{n,D} = \sum_{k=1}^n ((D+1)k - n) \binom{n}{k}^2 Q_{n-k,D}. \quad (20.1)$$

To verify it, write $F(t) = \sum f_k t^k$, $F(t)^D = \sum g_n t^n$, and compare coefficients in $F(F^D)' = DF'F^D$. This gives $ng_n = \sum_{k=1}^n ((D+1)k - n) f_k g_{n-k}$; multiplication by $(n!)^2$ proves (20.1). The closed-walk count is then the positive finite sum

$$A_{N,D}(\lambda) = \sum_{k=0}^{\lfloor N/2 \rfloor} \binom{N}{2k} \binom{2k}{k} Q_{k,D} \lambda^{N-2k}. \quad (20.2)$$

For rational λ , this is an exact rational computation. The arithmetic-operation count does not bound bit complexity: the integers can be large, and the recurrence itself includes signed terms.

The independent implementation multiplies truncated rational power series directly. A third implementation enumerates endpoints of short walks by dynamic programming, retaining separate

m	Cases	Leading	Through E_1	Through E_2
20	22	4.17×10^{-3}	4.199×10^{-5}	1.576×10^{-6}
40	22	2.085×10^{-3}	1.028×10^{-5}	1.924×10^{-7}
80	22	1.042×10^{-3}	2.545×10^{-6}	2.378×10^{-8}

Table 2: Maximum observed absolute relative error, with the exact count in the denominator, over the nonzero finite verification cases at each m . The grid includes both parities, sparse, balanced, dense, idle-only, and rational anisotropic cases. Higher-order columns are numerical diagnostics, not finite error certificates.

positive and negative directional weights. These methods check different parts of the combinatorial identity. The finite counterexample also has its own standard-library-only verifier, including the Machin lower bound for π and the exact positive integer comparison.

20.2 Verification coverage

The saved run used Python 3.12.14 and `mpmath` 1.3.0 at 60 decimal digits for numerical work. It passed the following 444 exact comparisons:

Comparison	Cases
Integer recurrence versus independent rational coefficients	91
Isotropic count versus direct endpoint enumeration	108
Isotropic count versus parity-reduced convolution	198
Asymmetric rational activities versus direct enumeration	36
Insertion or removal of inactive axes	11

The direct enumerations have dimension at most three and length at most eight. The coefficient comparisons extend through order twelve. These bounds describe the scope of the checks, not restrictions on the theorems. The dimension-16 counterexample is checked separately by two exact coefficient methods. The prescribed-endpoint corollary has an additional 1,303 exact comparisons in `code/verify_endpoints.py`: eight activity models in dimensions zero through two, lengths zero through six, rational asymmetric and one-sided axes, and all stated boundary types. This separate grid compares the coefficient formula with direct step enumeration for every reached endpoint and the box $\{-3, \dots, 3\}^D$.

There are 84 finite saddle cases at $m = 20, 40, 80$, covering both parities, sparse, balanced and dense parameters, zero and positive idle activities, idle-only walks, and rational anisotropy. Of these, 66 have nonzero counts and 18 are exact-zero boundaries. The program checks the saddle mean, $m/2 \leq B \leq m$, agreement of the universal coefficient generator with the displayed first two corrections, and the explicit $12/B$ envelope. The observed range of B/m is $[0.5, 0.993776729)$.

Table 2 uses the conventional error “approximation divided by exact count, minus one.” The rigorous envelope (16.11) uses the reverse normalization, “exact count divided by leading approximation, minus one.” Both values are stored in the data, and Figure 1 uses the latter normalization when comparing to $24/m$.

20.3 The dense and sparse transitions

For the dense illustration, D ranges through 100, 1600, 25600, 409600, and N is the even integer $sD^{3/2}$ or its odd successor, with $s = 1/2, 1, 2$. The 24 entries in Table 3 evaluate the parity saddle through the second correction. They are not exact enumerations at those large lengths. Their convergence to $\exp(1/(48s^2))$ illustrates Theorem 19.1; the theorem’s proof supplies the uniform

D	s	Even N	Odd N	$e^{1/(48s^2)}$
100	1/2	1.07791242	1.06895026	1.08690405
100	1	1.00426257	1.00417799	1.02105186
100	2	0.996040738	0.996039869	1.00522192
1600	1/2	1.08156499	1.08155921	1.08690405
1600	1	1.01680782	1.0168072	1.02105186
1600	2	1.00293032	1.00293026	1.00522192
25600	1/2	1.08550721	1.08550712	1.08690405
25600	1	1.01998889	1.01998888	1.02105186
25600	2	1.00464922	1.00464922	1.00522192
409600	1/2	1.08655137	1.08655136	1.08690405
409600	1	1.020786	1.020786	1.02105186
409600	2	1.00507876	1.00507876	1.00522192

Table 3: Dense transition: parity saddle approximations through E_2 , divided by the Gaussian prediction, at even $N = sD^{3/2}$ and the next odd integer. These large-dimension entries are saddle evaluations, not exact walk counts.

m	Even, leading	Even, corrected	Odd, leading	Odd, corrected
20	-1.916×10^{-2}	2.917×10^{-4}	-1.502×10^{-2}	8.558×10^{-5}
40	-9.749×10^{-3}	7.197×10^{-5}	-7.59×10^{-3}	2.093×10^{-5}
80	-4.917×10^{-3}	1.787×10^{-5}	-3.815×10^{-3}	5.169×10^{-6}
160	-2.469×10^{-3}	4.45×10^{-6}	-1.912×10^{-3}	1.284×10^{-6}

Table 4: Sparse count formula at $\theta = \beta = 1$: signed relative errors against exact normalized integer counts, before and after the $1/m$ correction. Here $D = m^2$ and $\lambda^2 = m$.

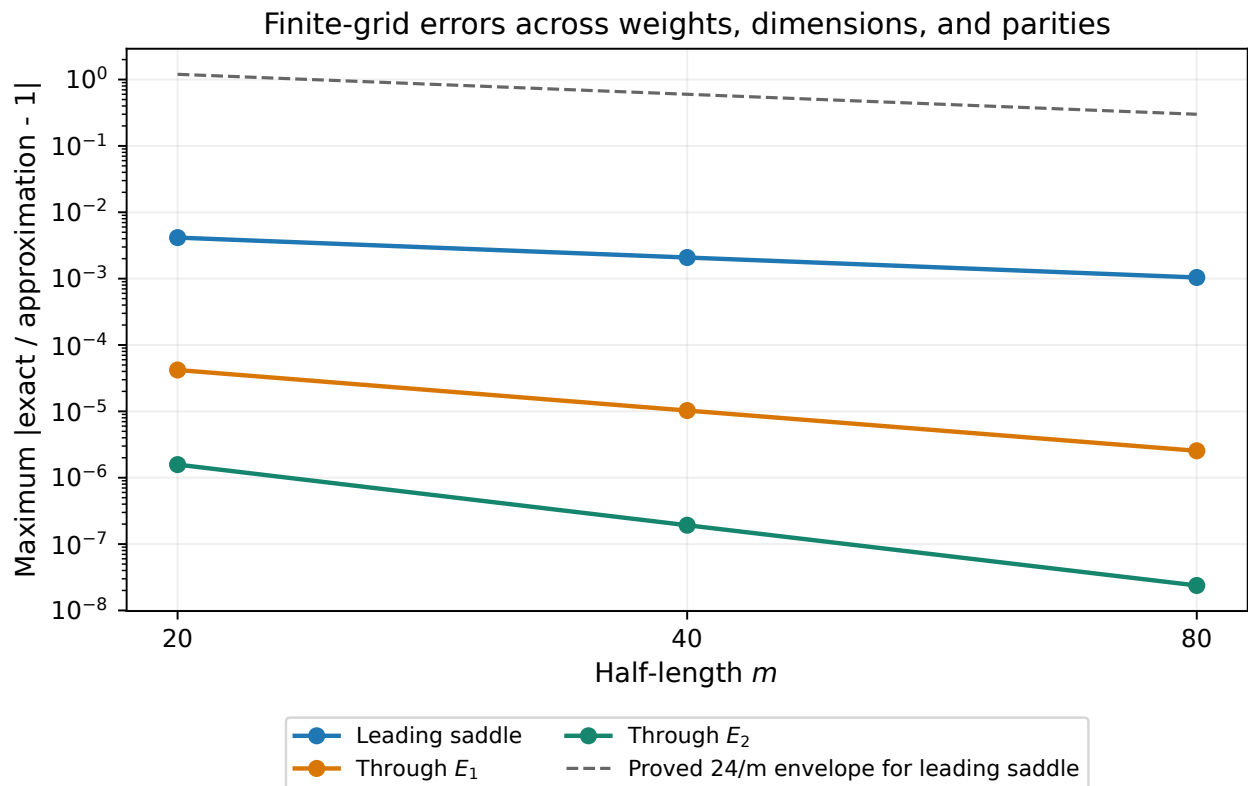


Figure 1: Finite exact-count comparisons for the uniform saddle expansion. The explicit $24/m$ leading envelope is conservative. The improved residuals after successive corrections are numerical observations; the all-orders theorem supplies their asymptotic orders.

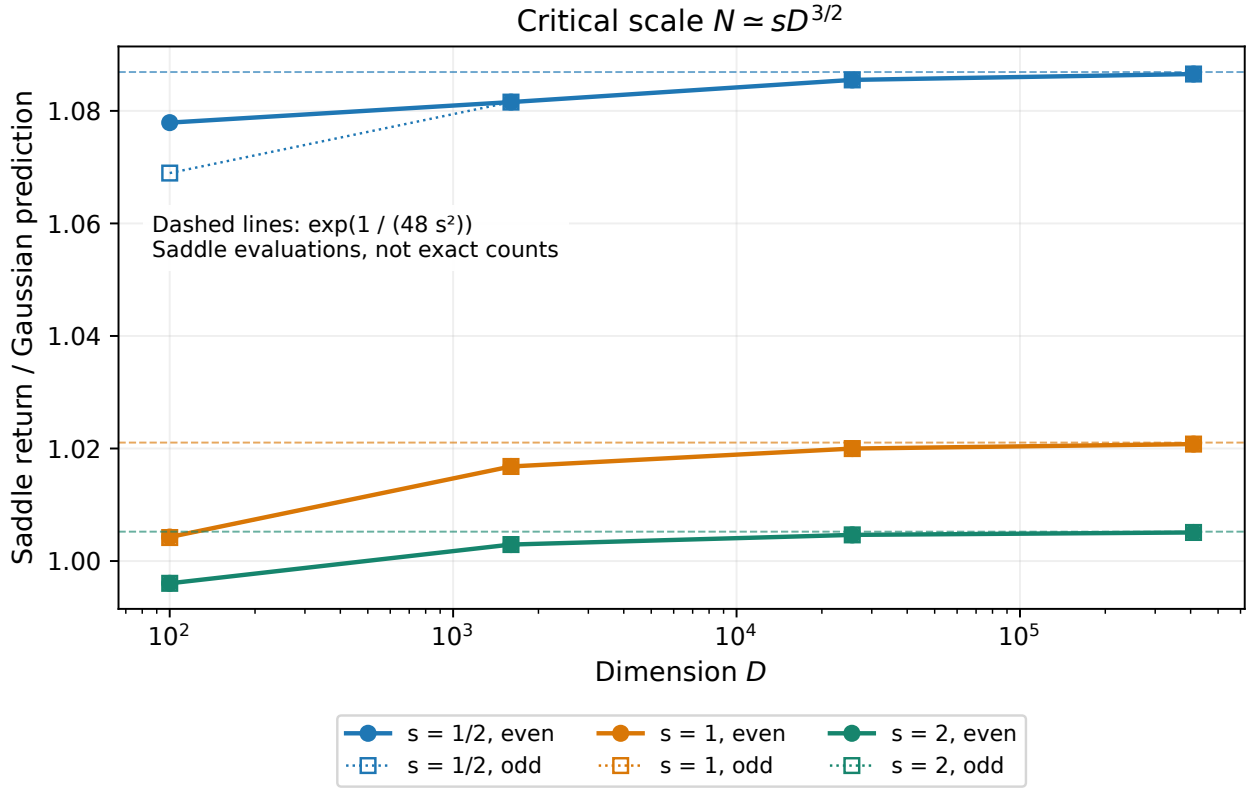


Figure 2: The dense Gaussian crossover. Saddle evaluations approach the limiting factor $\exp(1/(48s^2))$ as the dimension grows at fixed $s = N/D^{3/2}$. Finite values can approach the limit slowly because the first correction has order $D^{-1/2}$ on this scale.

justification. At moderate dimension, the negative $-3D/(16N)$ term and parity effects remain visible.

For the sparse illustration, $D = m^2$ and $\lambda^2 = m$, so $\theta = \beta = 1$. Removing the common factor λ^ϵ makes the count an exact integer, even when λ is irrational. Eight comparisons, for both parities and $m = 20, 40, 80, 160$, show the improvement from the correction in Theorem 18.1; see Table 4. This tests the counting correction independently of the global saddle expansion.

20.4 How to reproduce and extend the calculations

The archive contains the following entry points:

`code/verify_lyons.py`

Verifies the exact finite counterexample with rational arithmetic. It needs only the Python standard library and runs independently of the numerical package.

`code/verify_endpoints.py`

Checks the general endpoint coefficient identity against an independent step-by-step dynamic program. It uses only exact rational arithmetic and the Python standard library.

`code/walks.py`

Implements exact isotropic counts, rational anisotropic convolution, the scalar parity saddle, cumulants, and arbitrary fixed-order Edgeworth corrections. Numerical evaluations use

logarithms to avoid overflow; rational activities passed to the numerical routines should first be converted to `mpmath` values at the chosen precision.

`code/reproduce.py`

Runs the complete saved verification grid and regenerates `data/verification.json`, the three tables, and both figures.

From the archive root, the commands are

```
python code/verify_lyons.py
python code/verify_endpoints.py
python code/reproduce.py
latexmk -pdf article.tex
```

Package dependencies are recorded in `requirements.txt`; the PDF can be built directly from the included tables and figures without rerunning the numerical work.

The finite exact certificates and the analytic inequalities have different roles from floating-point diagnostics. The latter do not perform outward rounding. A certified numerical interval for a large count requires enclosing the saddle, its generator value, and its variance, then applying (16.11).

[write] **The shipped layout and the reruns, 7 October 2026.** This section describes the delivered package, which is not shipped as such. In this directory `code/verify_lyons.py`, `code/verify_endpoints.py`, `code/walks.py` and `code/reproduce.py` are `code/02-bridges-*.py`; `data/verification.json` and `data/numerical_tables.tex` are `data/02-bridges-*`; `requirements.txt` is `data/02-bridges-requirements.txt`; the figures are `figures/02-bridges-*`. `reproduce.py` imports `walks` by its delivered name and writes `data/` and `figures/` under the package root by the delivered names, overwriting the records there, and `latexmk -pdf article.tex` builds the manuscript, not this report; the report README gives a route that restores the delivered layout on a copy. Tables 2–4 are input from the shipped table file, unchanged. At the write, on a copy in the delivered layout: `verify_lyons.py` and `verify_endpoints.py` (standard library) passed, the latter with its 1,303 comparisons; `reproduce.py` (Python 3.13.5 under uv, mpmath 1.3.0, 85 s) passed its 444 exact checks, 84 saddle cases, 24 dense and 8 sparse cases, and wrote a `verification.json` equal to the shipped one except the recorded Python version (the shipped run used 3.12.14) and a table file equal to the shipped one apart from line ends.

21 Further research questions and a development programme

The results leave several concrete problems. The first three are the most direct continuations: they seek usable replacements for the disproved Gaussian bound, a precise transition window, and effective certification. Later items extend the probability laws or the class of walks. Each question is proposed as a task still open within this investigation; an exhaustive literature search remains necessary before assigning historical novelty.

21.1 A corrected universal Gaussian upper bound

The original conjecture cannot be repaired by changing its constant. The dense theorem suggests an exponential correction depending on D^3/N^2 . Determine the smallest explicit functions $U(D, N)$ for which

$$R_{N,D}^0 \leq 2G_{N,D}^0 e^{U(D,N)}$$

holds for every positive even N and $D \geq 1$, and for which $U(D, N) \sim D^3/(48N^2)$ whenever that term dominates in the dense range. A result restricted to $N \geq cD$ with an explicit c would already be useful. The positive scalar saddle and the $12/B$ local bound provide a rigorous starting inequality.

The remaining task is to replace the implicitly defined saddle by an explicit expression while retaining the sharp exponential scale.

Corollary 19.4 already determines the leading last-failure thresholds: $T_0(D) \sim D^2/12$ and $T_1(D) \sim D^2/9$. What remains is an explicit dimension onset for the uniform upper-bound region, lower-order terms in these thresholds, and a classification of finite-dimension exceptions. Those questions demand more information than the leading quadratic scale.

21.2 The narrow window near the quadratic crossing

For nonlazy walks the leading sign changes at $N/D^2 = 1/12$; for the particular lazy walk it changes at $1/9$. Determine the next terms in the crossing locations and the profile in windows such as

$$N = \frac{D^2}{12} + \alpha D + O(1)$$

for even nonlazy lengths. On this scale the leading terms already proved cancel, so a new term in the uniform scalar expansion and an explicit first saddle correction are required. It would also be valuable to prove uniqueness, or a controlled number, of crossings of the ratio through one at each sufficiently large dimension. The present sign corollary does not assert such uniqueness.

21.3 Effective high-order bounds and a verified evaluator

The $24/m$ envelope is explicit, whereas the constants C_K in the all-orders theorem are existential. Produce a recurrence for usable constants and an onset for each K . A practical implementation should enclose Bessel values and derivatives, locate the monotone saddle by interval arithmetic, and bound the local integral together with its tails. One can then compare the cost of using more terms with that of evaluating an exact finite coefficient.

A related sharp problem is to determine

$$\sup B \left| \sqrt{2\pi B} \mathbb{P}(S = \mathbb{E}S) - 1 \right|$$

over Bernoulli sums with integer mean, and the corresponding supremum over the narrower Bessel–hyperbolic family. The proof gives the upper bound 12 in the first class. Numerical improvement alone would not identify the extremal distributions or exclude singular parameter limits.

21.4 From rare collisions to Gaussian occupation fluctuations

The sparse joint theorem assumes $m^2/D = O(1)$. Investigate the intermediate sparse region

$$m/D \rightarrow 0, \quad m^2/D \rightarrow \infty.$$

Repeated axes now have a diverging mean. Determine the range in which a Poisson approximation remains accurate in total variation, and the range in which centering and scaling yields a normal law. The next structural scale is the appearance of a nonnegligible number of triple-use axes, measured at first order by m^3/D^2 . The marked logarithm in (18.12) identifies the terms that must be retained.

A successful theorem should control the joint law of J , the number of double-use axes, and the number of triple-use axes, and should match the proportional-dimension Gaussian occupation law in the predecessor [10]. This is a probability-law problem beyond the already uniform counting formula.

21.5 Anisotropic collision laws and effective dimension

The counting theorem permits arbitrary c_i , but the sparse law was proved only for $c_i = 1$. For a sparse anisotropic tilt, the quantities $\sum_i c_i$, $\sum_i c_i^2$, and $\sum_i c_i^3$ suggest the relevant pair- and triple-collision scales. Formulate a condition excluding domination by a single coordinate, then prove an anisotropic joint Poisson law with explicit error. The natural candidate effective dimension is

$$D_{\text{eff}} = \frac{(\sum_i c_i)^2}{\sum_i c_i^2},$$

when the denominator is nonzero. Determine how this parameter and higher normalized power sums enter the first correction. The formula is a proposed organizing parameter, not a theorem that one may substitute it for D everywhere.

21.6 A moving idle activity in the dense regime

The global theorem is uniform in λ , while the dense crossover was calculated for $\lambda = 1$ and $\lambda = 0$. Classify the dense asymptotics when the idle activity moves between these regimes or grows with D . The parity suppression is controlled by the size of λ times the positive saddle, so its transition can occur on a different scale from the Gaussian correction. A full theorem should identify when both lattice arcs matter, when one is negligible, and how the covariance normalization changes the first two uniform coefficients. The exact H_ϵ representation remains available throughout.

21.7 Nonzero endpoints and other step sets

Corollary 16.5 already gives the uniform coefficient expansion for arbitrary prescribed endpoints, using Bessel factors of orders $|x_i|$. Determine explicit Gaussian and large-deviation simplifications of that saddle when the coordinate profile of the endpoint moves with the dimension. In particular, characterize which power sums or maximal-coordinate conditions determine the transition between central and displaced regimes. Near ballistic endpoints the residual budget $\lfloor (N - |x|_1)/2 \rfloor$ may be small, and a separate exact or rare-event approximation may be preferable. The variance lower bound and the negative-zero representation themselves are already proved for these endpoints.

More generally, characterize finite symmetric step sets whose parity- or residue-reduced return generator has only negative real zeros, or factors into functions with an equally useful local probability representation. For a general graph or a step set with diagonal moves, coordinate independence can disappear; the present product argument does not automatically survive.

21.8 A staged formalization in ProveIt

The exact finite counterexample is the most accessible first target. Formalize the multinomial recurrence, compute $C_{13,16}$ by a verified evaluator, prove the rational Machin lower bound, and close the strict integer inequality. This yields an independently checkable formal theorem without first developing asymptotic analysis.

The analytic programme can then be separated into reusable results: countable Bernoulli products and coefficient tilting; the uniform characteristic-function bound; the explicit local inequality; the Bessel product and variance estimate; and the finite-order integrated expansion. The sparse coefficient-transfer lemma is another short independent target. The dense crossing theorem would follow only after a formal uniform remainder for the differentiated Bessel expansion. This order gives useful intermediate theorems at each stage.

21.9 Implications for spanning-tree entropy estimates

The return-probability conjecture arose in the study of spanning-tree entropy, where Felker and Lyons express the entropy through a series of return probabilities [11]. Investigate tail bounds that combine exact initial coefficients with the corrected dense estimate. The finite counterexample by itself does not invalidate results whose tails use a separately established large-time bound; the threshold and the assumptions of each application must be checked. A useful outcome would be dimension-uniform certified truncation rules for this series.

Recommended order of work. First formalize the exact counterexample and build an interval evaluator for the already proved saddle bound. Then derive a corrected explicit Gaussian upper inequality and resolve the quadratic crossing window. The marked sparse transition and anisotropic probability laws provide the next coherent research programme beyond counting.

[write] **Under Vladimir’s standing rule of 4 October 2026, 7 October 2026.** The nine subsections above are questions, and they stay open; the leading thresholds of Section 21.1 are proved (Corollary 19.4) and only their onsets and lower-order terms are asked. The write found no claim of Part II that is unproved as stated outside these questions and none that is wrong, so nothing was moved here or refuted. Recorded by dated notes: Ball–Sterbenz state their display for $n \geq 0$ and Felker–Lyons display their (2.3) with a qualification (end of Section 15); the Gowers–Szemerédi ledger count is stale since the pin (end of Section 14); the delivered layout (Section 20). Two of these questions meet Part I’s: Section 21.3 is Part I’s Question 1 for Part II’s expansion, and Section 21.4 asks for the laws between Part II’s sparse range and Part I’s compact range, the part of Part I’s Questions 2 and 3 left open on the sparse side (note at the end of Section 12). On Section 21.8: no Lean or Rocq declaration exists for any statement of this report. On Section 21.9: Felker and Lyons’s rigorous bounds use the inequality only in dimensions 3 to 6 (note at the end of Section 15).

References

- [1] OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*, A328716. Exact formula attributed to Ilya Gutkovskiy (26 October 2019); leading equivalent and numerical constants attributed to Václav Kotěšovec (27 October 2019). [Entry](#); [official source record](#). Inspected 3 October 2026.
- [2] OEIS Foundation Inc., A328718, square array of lazy closed lattice walks. Entry by Seiichi Manyama (26 October 2019); walk interpretation by Alois P. Heinz (26 October 2019). <https://oeis.org/A328718>. Inspected 3 October 2026.
- [3] I. Gessel, J. Weinstein, and H. S. Wilf, *Lattice walks in $\mathbb{Z}^d$ and permutations with no long ascending subsequences*, *Electronic Journal of Combinatorics* **5** (1998), Research Paper 2, 11 pp. [Primary journal PDF](#), especially equation (2), pp. 2–3. [Part II cites it for background, with the article page <https://www.combinatorics.org/ojs/index.php/eljc/article/view/v5i1r2>: “See especially the Bessel generating function in equation (2).”]
- [4] P. Flajolet and R. Sedgewick, *Analytic Combinatorics*, Cambridge University Press, 2009. Theorem VIII.8 and proof, pp. 587–588; remarks and Note VIII.37, p. 589. [Author-hosted book PDF](#).
- [5] V. Reshetnikov, *ProveIt*, GitHub repository, inspected at commit 136e707021c9, 7 October 2026. [Pinned repository snapshot](#). [Part II.]
- [6] V. Reshetnikov, *math*, GitHub repository, inspected at commit adc7f1241b42, 7 October 2026. [Pinned repository snapshot](#). Catalogue and README counts refer to this snapshot. [Part II.]

- [7] *math*, formalization scope for family 159, [lean/docs/159.md](#), at the snapshot in [6]. [Pinned scope document](#). [Part II.]
- [8] *math*, formalization scope for family 241, [lean/docs/241.md](#), at the snapshot in [6]. [Pinned scope document](#). [Part II.]
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